

BrickAtlas: Supplementary Evidence

Development Manuscript
Author metadata intentionally omitted

1. Benchmark Reference

This supplement separates three types of evidence: measured full-casebank audits, measured adaptive model studies, and explicitly unmeasured future experiments. The latter have blank result cells and must not be interpreted as zero accuracy. All new figures are generated from archived renders or machine-readable evidence, not illustrative model outputs.

1.1. Curated Calibration Cases

The 5,120-source v2 population remains unchanged and preserves all historical scores. A new `curated-casebank-1` layer provides twelve original, explicitly authored cases for qualitative inspection, task calibration, and future human studies. It is not counted as 5,132 v2 sources and is not a hidden confirmation set.

Every structure passes both geometry implementations and a piece-by-piece vertical insertion check. The examples span furniture (bench, picnic table), architecture (staircase, gate, pavilion, townhouse, station), infrastructure (single- and double-span bridges), and landmarks (watchtower, lighthouse, two-tier pagoda). They use two to four role-based colors rather than independently sampled part colors. “Hard” may denote a long dependency chain or multiple separated supports rather than only more pieces.

Each model directory includes `structure.json`, `metadata.json`, four views, every occupied layer, an exploded view, and legal assembly checkpoints. Each task directory includes the public task contract, expected output, input renders where applicable, and ground-truth renders. `CONTACT_SHEET.png`, `TASK_GROUND_TRUTH.png`, and `index.html` provide direct review entry points. The designs are original CC0 BrickAtlas assets; no official-set geometry is copied. Their semantic recognizability remains a claim to validate with independent people.

1.2. Advanced Challenge Layer

The measured v2 study exposes two opposite regimes: one-clause recoloring and removal are saturated, while ordinary reconstruction, completion, and shifted-pose repair often remain at zero exact success. Moreover, a height-sorted algorithm solves all v2 planning cases. The separate

`challenge-casebank-2` layer therefore changes both scale and task structure without modifying v2 or historical scores. Revision 1 is archived; revision 2 corrects public candidate poses, strict types, yaw equivalence and the inspection observation loop.

The contracts are recovery planning from a legal dead end, iterative support counterfactuals, a three-clause compound edit, simultaneous color/pose/extra-part repair, costed active inspection, shortest-path and articulation reasoning, non-suffix distributed completion, rigid subassembly translation, legal-next-part set prediction, selected-part pose estimation, 53-part reconstruction, and inventory-constrained redesign. Recovery actions are executed in submitted order. The counterfactual repeatedly removes newly unsupported parts. The selection oracle returns all legal candidates, not one arbitrary next piece. Full-target visual tasks now add bottom-layer disclosure. Multi-fault repair requires both exact restoration and the complete set of wrong current IDs.

This layer remains far smaller than BrickNet and lacks LDraw connector diversity, force stability, manuals, and robot actions. Its purpose is controlled cross-capability stress testing, not a claim of comparable asset scale or generative realism.

1.3. Source Policies and Catalog

Figure 5 shows all eight construction procedures. Table 4 gives their measured means; height uses plate units, and multi-supported counts pieces with at least two distinct lower supporting parts. Each row contains 640 source objects. These differences are useful geometric controls but not eight independently collected semantic categories. The shared root plate makes retrieval and prior-shape controls especially important.

1.4. Construction and Expansion

For each policy, candidate seeds are deterministic. The pipeline is:

1. Generate a valid insertion sequence from the common root plate under the policy-specific attachment rule.
2. Normalize horizontal origin; reject invalid sizes and exact colorless translation/yaw-equivalent duplicates, including old releases.

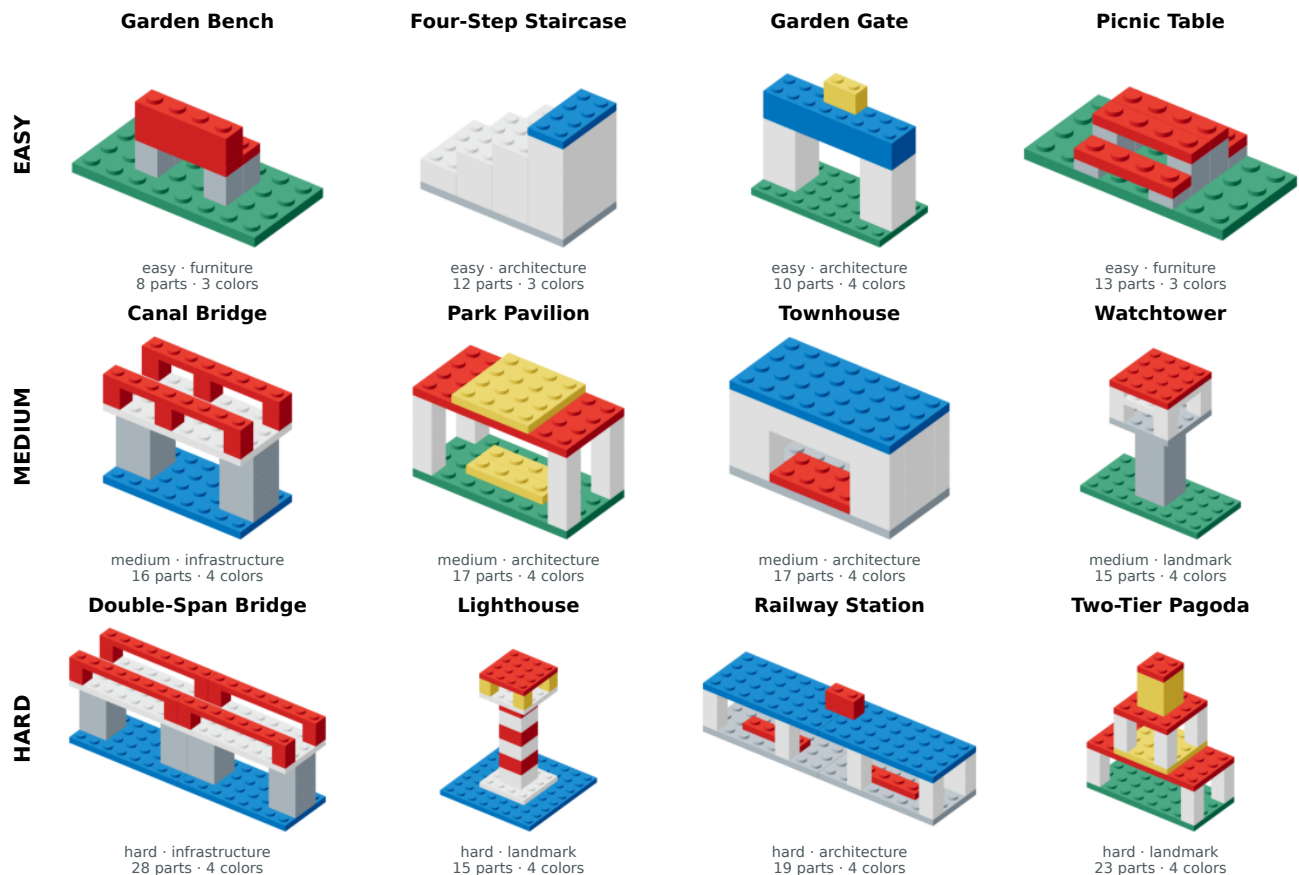


Figure 1. The complete curated set: four easy, four medium, and four hard structures. Difficulty combines part count, occupied levels, occlusion, support structure, and span length. Colors encode functional roles and never exceed four per object.

Model	Difficulty	Style	Parts	Colors	Levels	Occlusion / support
Garden Bench	easy	furniture	8	3	4	low / single
Four-Step Staircase	easy	architecture	12	3	6	low / single
Garden Gate	easy	architecture	10	4	7	low / multiple
Picnic Table	easy	furniture	13	3	4	low / multiple
Canal Bridge	medium	infrastructure	16	4	7	medium / spanning
Park Pavilion	medium	architecture	17	4	6	medium / multiple
Townhouse	medium	architecture	17	4	5	high / multiple
Watchtower	medium	landmark	15	4	10	medium / multiple
Double-Span Bridge	hard	infrastructure	28	4	7	medium / spanning
Lighthouse	hard	landmark	15	4	11	medium / multiple
Railway Station	hard	architecture	19	4	5	high / multiple
Two-Tier Pagoda	hard	landmark	23	4	11	high / multiple

Table 1. Curated-case specification. These labels describe explicit design contracts, not measured human difficulty. Exact structures, palettes, legal orders, views, layers, and steps are stored under `benchmark/curated-cases/`.

3. Run the independent integer-cell geometry check. Accept until 640 unique geometries are available per policy.
4. Assign the source split from the geometry hash, except for the two entirely held-out policies.
5. Derive task variants and reference conditions; attach the source group to every derivative; build the public input and evaluator record separately.
6. Validate oracle success, empty-output failure, metric ap-

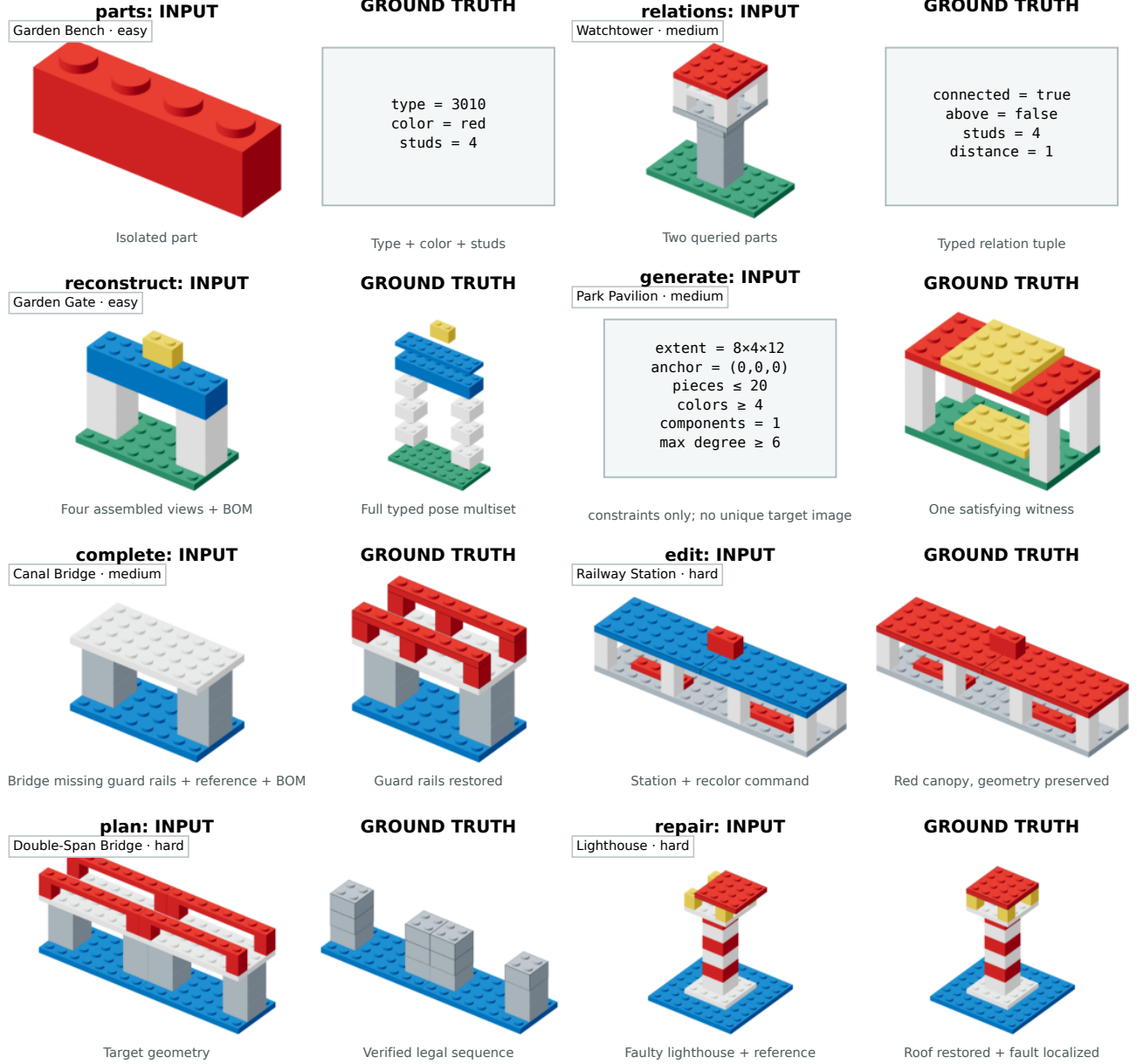


Figure 2. All eight curated task demonstrations. Text panels show non-image labels or constraints; visual panels show actual input structures and evaluator ground truth. Generation has no unique target, so the pavilion is explicitly one satisfying witness. Planning ground truth is a verified prefix, not merely a final object render.

plicability, render recipes, and hashes before materializing compressed shards.

The release contains 14 shards and 476,760 serialized rows across source, input, ground-truth, render, and case-index records. Those storage rows are not extra samples.

1.5. Input and Answer Schemas

The public exporter emits `caseId`, a fixed system prompt, the public input task, and relative images paths.

The task contains a question, public structure/constraints where applicable, and a response schema. The full evaluator index is public for reproducibility; ordinary-input isolation is a benchmark protocol, not a security guarantee against consulting the answers.

The following is a minimal *schema example*, not a model result:

```
{"version":1,"parts":[
  {"id":"p0","partId":"3005",
```

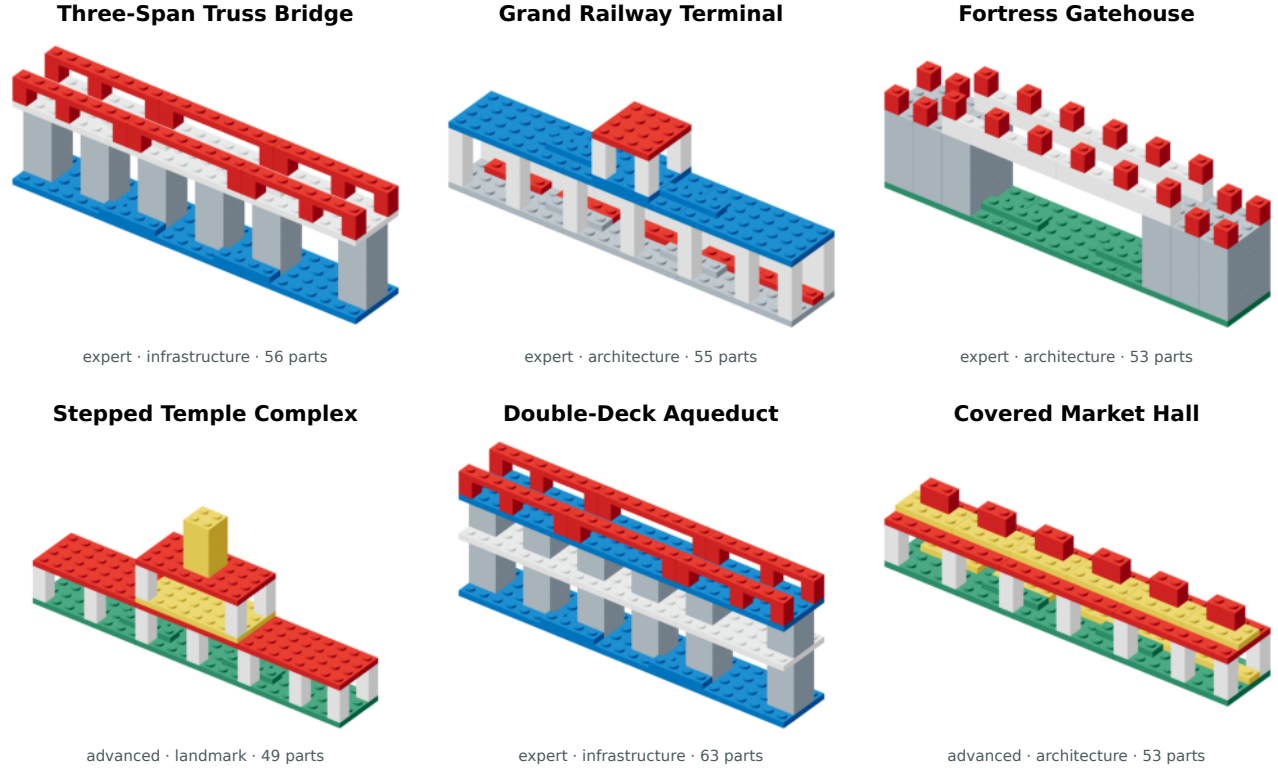


Figure 3. Six original challenge structures. They contain 49–63 parts each (329 placed instances total), compared with 8–28 parts in the curated calibration layer. All are connected and have verified piece-wise vertical insertion orders.

Model	Tier	Domain	Parts	Intended pressure
Three-Span Truss Bridge	expert	infrastructure	56	56-part long-horizon plan; Multiple legal frontiers; Recoverable deck-first dead end
Grand Railway Terminal	expert	architecture	55	55-part scene reconstruction; Repeated-instance disambiguation; Compound regional edit
Fortress Gatehouse	expert	architecture	53	53-part occlusion; Long graph paths; Inspection choice under hidden structure
Stepped Temple Complex	advanced	landmark	49	49-part distributed completion; Nested subassembly transform; Strong symmetry traps
Double-Deck Aqueduct	expert	infrastructure	63	Fine-grained next-part selection; Pose estimation among repeated bays; 63-part output length
Covered Market Hall	advanced	architecture	53	53-part multi-view reconstruction; Inventory-constrained redesign; Repeated-part correspondence

Table 2. Challenge structures and design intent. Tiers are controlled design strata, not human-calibrated difficulty labels.

```

"color": "blue", "x": 0, "y": 0,
"z": 0, "turn": 0}
]]

```

Recognition returns `partId`, `color`, `studs`. Relations return `connected`, `above`, `contactStuds`, `shortestPath`. Plans return `order`, an array of all target IDs exactly once. Repair wraps a full program as structure and adds `faultIds`; no-fault answers use

an empty array. Normal reconstruction, generation, completion, and editing return a structure.

Relative nodes may instead specify an earlier `parent`, an offset in global grid axes, and absolute `turn`. This encoding is reversible but does not assert a stud connection; it is not BrickNet’s typed connector language. Square footprints admit all quarter-turns, while other rectangular footprints admit half-turn symmetry. Scoring ignores instance IDs for multiset recovery but does not globally align transla-

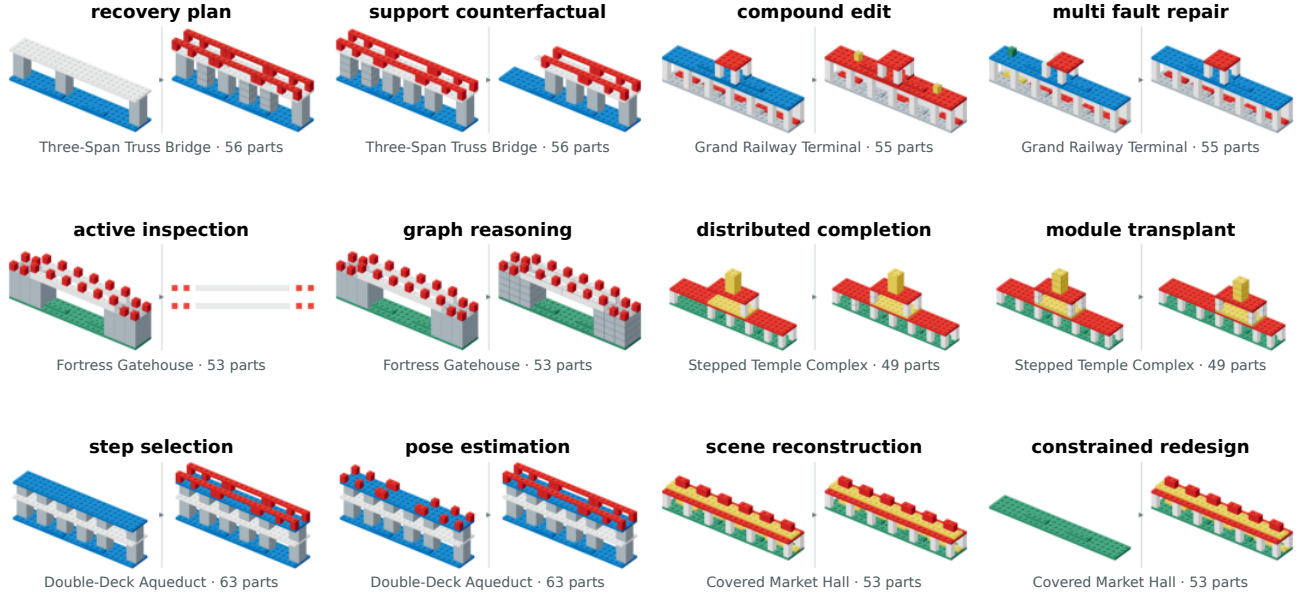


Figure 4. Twelve advanced contracts, each shown as input/context (left) and oracle target or disclosed consequence (right). Visual targets do not replace the typed JSON oracle. Constrained redesign accepts any satisfying structure rather than only the displayed witness.

Task	Source	Parts	Model result
recovery-plan	Three-Span Truss Bridge	56	
support-counterfactual	Three-Span Truss Bridge	56	
compound-edit	Grand Railway Terminal	55	
multi-fault-repair	Grand Railway Terminal	55	
active-inspection	Fortress Gatehouse	53	
graph-reasoning	Fortress Gatehouse	53	
distributed-completion	Stepped Temple Complex	49	
module-transplant	Stepped Temple Complex	49	
step-selection	Double-Deck Aqueduct	63	
pose-estimation	Double-Deck Aqueduct	63	
scene-reconstruction	Covered Market Hall	53	
constrained-redesign	Covered Market Hall	53	

Table 3. **Unmeasured.** The challenge task list. Blank cells are deliberate: oracle and near-miss validation are complete, but no model has been run on these new cases.

tions or rotations unless the task explicitly requests a transformation.

1.6. Render and Coordinate Contract

The released views are 640-by-480 PNGs. X, Z are stud coordinates; Y is bottom plate height. The front camera looks from positive Z , with image-right $+X$ and image-up $+Y$. The side camera looks from positive X , with image-right $-Z$ and image-up $+Y$. Top looks from positive Y with a small positive- Z offset, image-right $+X$ and image-down $+Z$. Cameras fit each structure; apparent pixel scale is not a fixed metric calibration. Layer views remove every part not starting at the selected bottom Y . They are additional

observations, not ordinary views through an opaque object.

1.7. Full No-Operation Audit

`study/paper-audit.ts` first verifies frozen casebank source hashes and the compressed structure shard digest, computes a per-source inventory, and evaluates the copy predictor on 35,840 selected cases. The predictor receives only the public task and copies `current`; repair adds an empty fault list. Prediction production does not inspect the evaluator target, changed IDs, or oracle. Its fixed output is then scored with the versioned strict evaluator.

Seven endpoints each contain one case per source: ordinary suffix completion, three default symbolic ed-

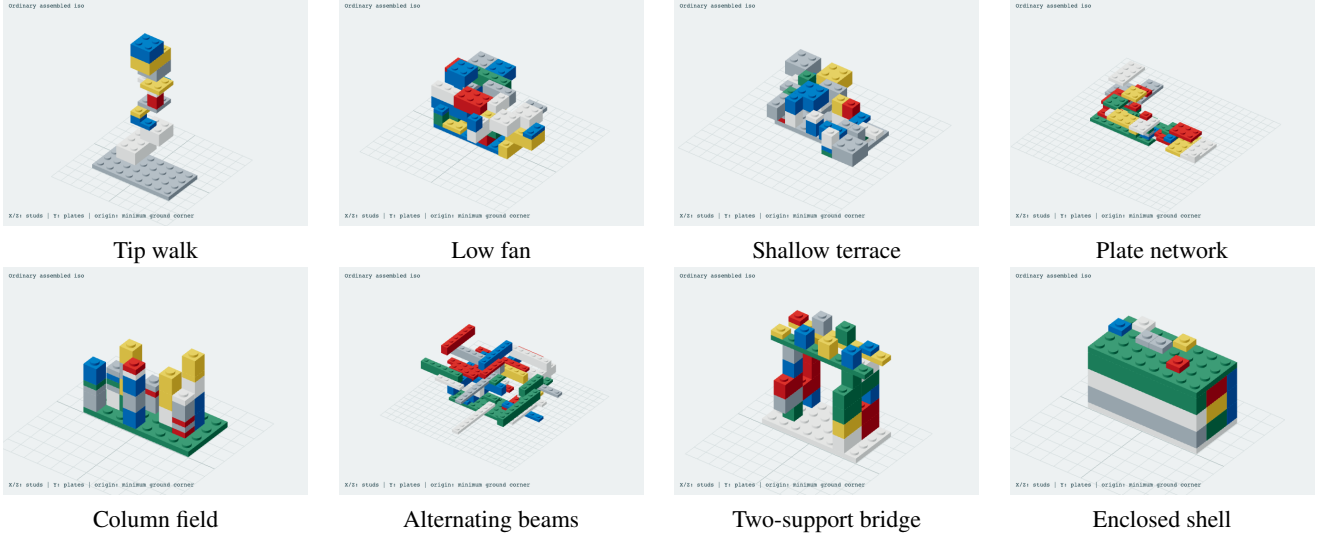


Figure 5. Actual largest-case-per-policy render-audit examples. Bridge and shell are policy-held-out. Selection stresses rendering complexity rather than providing a random visual summary.

Policy	N	Parts	Height	Multi-sup.
tip-walk	640	11.3	25.7	0.00
low-fan	640	25.0	12.4	0.75
shallow-terrace	640	25.0	6.9	0.30
plate-network	640	25.0	8.3	1.64
column-field	640	23.8	11.9	0.00
alternating-beams	640	25.1	16.4	3.29
two-support-bridge	640	25.0	17.3	2.00
enclosed-shell	640	25.9	9.0	1.00

Table 4. Full-release geometric means. Multi-sup. counts multiply supported parts per object, not a physical-stability score.

its, and ordinary no-fault/color/shift repair. Raw case-level scores, IDs, policies, and splits are retained in `copy-controls.jsonl.gz`; selection and artifact hashes are recorded in `audit.json`. Summary code rejects duplicate cases or repeated sources within an endpoint and records metric-specific denominators. These are descriptive census values on the public release, not sampling intervals or held-out model performance.

2. Additional Measured V2 Results

2.1. Coordinate-Origin Revision

The original 24-call nested 2/4/8-piece ladder retained minimum $Z = 1$ for two prefixes despite inherited minimum-corner wording. One failed response matched up to translation. Original outputs and scores are preserved. A separate zero-anchored, re-rendered 24-call revision has RGB+BOM success 1/12 and symbolic success 11/12. Both ladders derive from one source, not six independent scenes.

ID	Catalog type	Instances	Objects
2456	Brick 2x6	0	0
3001	Brick 2x4	1,799	948
3002	Brick 2x3	1,845	963
3003	Brick 2x2	3,859	1,573
3004	Brick 1x2	11,736	2,889
3005	Brick 1x1	28,210	3,540
3007	Brick 2x8	0	0
3008	Brick 1x8	4,805	1,250
3009	Brick 1x6	2,568	611
3010	Brick 1x4	4,865	1,581
3020	Plate 2x4	1,836	584
3021	Plate 2x3	1,991	595
3022	Plate 2x2	6,162	2,166
3023	Plate 1x2	7,221	2,209
3024	Plate 1x1	20,957	4,153
3031	Plate 4x4	0	0
3032	Plate 4x6	0	0
3034	Plate 2x8	0	0
3035	Plate 4x8	5,760	5,120
3460	Plate 1x8	3,528	1,247
3622	Brick 1x3	2,151	985
3623	Plate 1x3	2,217	605
3666	Plate 1x6	2,645	617
3710	Plate 1x4	4,915	1,222
3795	Plate 2x6	0	0

Table 5. Exact catalog coverage in the 5,120 source assemblies. Instances count placements; objects count sources containing a type. Zero-frequency types remain in the 25-type recognition catalog.

Origin, wording, image rendering, and re-sampling change together, so the difference cannot be assigned solely to ori-

Task / variant	GPT-4.1 mini	Gemini Flash
parts	8/25	12/25
relations/contact	7/24	7/24
relations/separated	0/24	0/24
reconstruct/full	0/24	0/24
generate/constraints	0/24	0/24
complete/suffix	0/24	0/24
edit/rotate	0/24	0/24
edit/recolor	24/24	24/24
edit/remove	24/24	24/24
plan/assemble	1/24	12/24
plan/disassemble	1/24	7/24
repair/none	21/24	17/24
repair/color	4/24	6/24
repair/shift	0/24	0/24

Table 6. The earlier 24-source v2 study, retained separately from the expanded three-source matrix. Recognition uses 25 catalog types. All structural variants share the same 24 source groups.

Copy-current task	Cases	Success	Part F1	Local
complete/suffix	5,120	0.0	87.1	0.0
edit/rotate	5,120	0.0	0.6	0.0
edit/recolor	5,120	0.0	79.0	0.0
edit/remove	5,120	0.0	97.4	0.0
repair/none	5,120	100.0	100.0	100.0
repair/color	5,120	0.0	94.9	0.0
repair/shift	5,120	0.0	94.9	0.0

Table 7. Full-casebank no-operation census. Local is addition F1, edit-target success, or repair-localization F1, respectively. Rows share 5,120 sources; this is not a learned-model test split.

Direction	Split	Cases	Successes	%
assemble	train	2,690	1,615	60.0
assemble	validation	392	230	58.7
assemble	test_id	758	468	61.7
assemble	test_ood	1,280	51	4.0
disassemble	train	2,690	2,690	100.0
disassemble	validation	392	392	100.0
disassemble	test_id	758	758	100.0
disassemble	test_ood	1,280	1,280	100.0

Table 8. Public-input greedy planning, original submitted order. Height sorting separately solves all 10,240 directional cases. Neither algorithm is a learned model.

gin normalization.

3. Earlier Neural Pilots

Two preceding releases provide neural pipeline evidence but not a v2 leaderboard. A 576-object composition pilot evaluated four held-out objects with GPT-4.1 mini and

Gemini control	Original	Control
no-image	0/8	0/8
mismatched-images	0/8	0/8
symbolic-reference	0/8	8/8
reflection	0/8	0/8
validation	0/8	0/8
repeat	0/8	0/8
relative	6/8	0/8

Table 9. Eight-source paired controls. Relative is assembly representation; the remaining arms are reconstruction. The symbolic arm adds privileged geometry.

Condition	Seed	Steps	Answer tokens	Success
base	17	0	0	0/337
leave-edit	17	120	61,440	7/337
leave-edit	29	120	61,440	8/337
leave-edit	43	120	61,440	9/337
multi	17	120	61,440	9/337
multi	29	120	61,440	2/337
multi	43	120	61,440	9/337
single	17	120	61,440	0/337
single	29	120	61,440	0/337
single	43	120	61,440	0/337

Table 10. All ten measured local v2 VLM jobs, 337 cases each. The nine trained runs match supervised answer tokens, not total image/input tokens, compute, or elapsed time.

Condition	Mini	GPT-4.1	Flash	Qwen
Full structure (RGB)	2/4	4/4	2/4	0/4
Local pose (RGB)	3/4	2/4	2/4	1/4
Choice (RGB)	2/4	2/4	2/4	4/4
Choice (permuted)	2/4	2/4	2/4	2/4
Choice (no image)	2/4	1/4	1/4	1/4
Local pose (symbolic)	4/4	4/4	4/4	4/4

Table 11. Measured four-source adaptive probes. Candidate lists, local-pose outputs, and symbolic references are different protocols, not interchangeable scores.

Gemini 2.5 Flash. Each model answered 56 cases, for 112 API requests. Table 12 shows visual task counts; all cells have only four objects. Both models failed ordinary and layer reconstruction/completion, but succeeded with the privileged symbolic reference. This suggests a useful perception-versus-structure diagnostic, not a statistically established model comparison. V2 also changes some scoring rules, including full-structure requirements under layer disclosure; cross-version scores must not be pooled.

A further Gemini control reuses the identical first answer and input for 12 ordinary-condition generation/reconstruction/repair cases. One additional call either

Task	Model	RGB	Layers	Symbolic
reconstruct	GPT-4.1 mini	0/4	0/4	4/4
reconstruct	Gemini Flash	0/4	0/4	4/4
complete	GPT-4.1 mini	0/4	0/4	4/4
complete	Gemini Flash	0/4	0/4	4/4
repair	GPT-4.1 mini	0/4	1/4	3/4
repair	Gemini Flash	2/4	0/4	3/4

Table 12. Earlier composition-v1 pilot only. Gemini Flash denotes Gemini 2.5 Flash. These 112 API calls are not v2 model evaluation.

Condition	Seed	Steps	Tokens	Success
base	17	0	0	0/8
single	17	120	195,292	2/8
multi	17	120	173,957	1/8
multi	29	120	176,765	1/8
multi	43	120	173,137	2/8

Table 13. Earlier Grid-v1 local text/structure pilot, eight test cases per row. Tokens count processed input plus output training tokens. Base/single have one seed each. This is not v2 or multimodal training.

requests reflection without new scene information or supplies target-blind legality feedback. Both arms remain at 2/12 exact successes, with no gained or lost successes. The 24 new requests are billed once; reused historical receipts are not charged again. Arms ran sequentially rather than interleaved, and the sample is too small to establish that validation tools are generally ineffective.

An older Grid-v1 text/structure pilot uses Qwen3-0.6B and rank-8 LoRA on query/value projections, with 1,146,880 trainable parameters, AdamW learning rate 2×10^{-4} , and 120 optimizer steps per trained run. The maximum sequence length is 3,072; overlength training examples are excluded rather than truncating answers. Greedy inference allows 1,200 new tokens. Single-task training uses editing; multi-task training uses relations, generation, editing, and planning. Table 13 reports actual outputs. All successes are editing cases. Multi-task seeds average 16.7% with a sample standard deviation of 7.2 percentage points, without evidence of broad transfer.

The single/multi runs are step-matched, not token-matched, and use different supervision diversity. A lower validation NLL is not evidence of execution success. Four adapters, five prediction sets, and all 480 optimizer-step records are bundled. Training used an Apple M4 Pro with 48 GB memory and MPS; local computation consumed real resources despite having no API bill.

4. Evidence Boundaries

The six-work-package research contract separates positioning, data, measurement, model experiments, independent review, and delivery. Same-machine runtime-free replay is not independent reproduction. Review packets contain no completed human judgments until real reviewers submit version-matched labels; disagreement is not silently adjudicated. No human-designed dataset is treated as imported merely because a paper or connector annotation is publicly accessible. Permission, assets and redistribution rights are separate gates.

5. V2 Training And Verification

All 117,910 strict-decoder oracle answers pass, 5,120 malformed primitive-type examples are rejected, and scores on 118,640 existing baseline/API records agree with their historical values. Independent implementations agree on 138,240 structural metric comparisons, all 10,240 relation queries, and connector counts on 5,120 structures. An additional 107,520 mutation/invariance checks cover missing, duplicate, unknown, and reversed plan actions, no-op edits, faulty localization, IDs, and yaw symmetry. Finite tests are not a proof over arbitrary malformed submissions, and shared catalog assumptions limit algorithmic independence.

The local pilot uses SmolVLM-256M-Instruct revision 7e3e67ed, unquantized. Ninety-six training and 24 disjoint validation objects provide seven task families (672/168 examples) and actual PNG views. Language-side query/value rank-8 LoRA has 460,800 trainable parameters. Single-reconstruction, multi-task and leave-edit use seeds 17, 29 and 43. Every trained run uses 120 updates, exactly 512 supervised answer tokens per update, AdamW at 2×10^{-4} , and gradient clipping at 1. Full prompts and answers are retained; deterministic answer-token subsampling matches supervision, not input/image tokens, time or compute.

Validation NLL uses two fixed examples per task. Training uses PyTorch/MPS; formal inference uses MLX with a 2,200-token cap on the same 337 cases. Token, pixel and mask digests agree across preprocessing backends, but floating-point kernels need not produce identical outputs. All 840 training/validation prompt prefixes retain token identity when answers are appended; maximum complete length is 5,176 tokens. Compact test-oracle answers use at most 1,793 tokens. Oracle lengths are private diagnostics, never appended to model inputs. The completed matrix retains 3,370 predictions, 1,080 updates and 552,960 supervised tokens. Each of nine adapters has 60 adapted matrices and 411 unchanged tensors independently checked under the actual bfloat16 cast-before-add merge semantics.

The calibration pool contains 468 questions from 36 validation objects excluded from the original study and

training/validation bundle. Selection cycles through available size bins within each of six policies; missing bins remain explicit. Portable inputs contain 720 image uses and 144 PNGs checked against public input, hashes and non-blank rendering. The earlier train/validation/test render audit checks 1,177 records, 1,970 image uses and 626 regenerated images on the same browser/GPU stack, not arbitrary rasterizers.

The casebank verifier checks 476,760 serialized rows across 14 shards, regenerates every ground truth and reconstructs baseline state chains. Browser checks cover desktop and mobile, nonblank scenes, camera interaction, input/truth isolation and export. Additional development responses are replayed offline and are not retroactively included in older browser-audit counts. Reproduction shares pinned Node dependencies; it is not an independently installed operating system or a second person’s training experiment.

6. Candidate Controls

For the initial pose probes, all candidate structures are legal, use identical part identities, yaw and height, and differ in horizontal position. Correct labels are balanced across the four selected scenes. Repeating a question and permuting its candidates are distinct interventions. The follow-up contract compares permutation mismatch to same-order repeat mismatch within each source/model, with temporal order balanced across source groups. Source-level bootstrap intervals describe only this selected development sample; they do not establish population coverage or stable model rankings.

6.1. Renderer-Aware Analysis by Synthesis

To test whether candidate locations contain usable image signal, a post-hoc deterministic baseline consumes only public BOM, candidate poses and RGB views. For the two-piece blue-on-gray domain, the supplied positive query height and support rules imply a grounded gray piece. Its horizontal positions and both distinct yaw footprints are enumerated under minimum-corner and contact constraints, not obtained from ground truth. Every legal candidate scene is rendered with the public renderer.

We threshold blue chroma ($B > 80$, $B > 1.25R$, $B > 1.1G$) inside image rows 32–447, excluding title/footer and neutral grid/text. A candidate receives the smallest mean mask-IoU distance over its legal hypotheses in top, front and side views. Equal candidate scores or missing blue masks cause abstention. No-image inputs abstain without rendering. The thresholds and enumeration are fixed in code before this run, but the method was designed after the preceding development experiments.

All 44 image-conditioned responses succeed, representing only 16 original sources with repeated/permutated con-

Development condition	Success	Abstain	Sources
Pose/choice-rgb	4/4	0	4
Pose/choice-permuted	4/4	0	4
Pose/choice-no-image	0/4	4	4
Order/original	12/12	0	12
Order/repeat	12/12	0	12
Order/permuted	12/12	0	12
Order/no-image	0/12	12	12

Table 14. Specialized known-renderer baseline. The 60 conditions reuse only 16 sources. No-image zeros are deliberate abstentions, not random-guess accuracy. This is not an additional neural model.

ditions. All 16 permutations preserve the selected pose. Cyclic replacement with the next source’s views causes one abstention and changes 9/15 jointly answered selections; these mismatched BOM/image inputs test dependence on pixels, not valid-task accuracy. The implementation has no calibrated out-of-domain rejection threshold for nonblank inputs. The run generates 384 candidate views in 29.7 seconds on the local browser stack, including 60 predictions and 16 image replacements, with no API calls. Cached templates enable offline rescoring.

Using the exact renderer, canonical colors and candidate information strongly favors this method. It is an algorithmic simulator-aware control, not a general reconstruction system, novel learned model, independent human validation or confirmation on new sources. Original VLM prompts also contained an isometric view, which this baseline does not use. Prediction production precedes loading evaluator targets for scoring.

7. Resource Accounting

The earlier 363-call campaign cost \$0.45102511, and the 730-call v2 study cost \$1.50635072. The four-model screen and both ladder versions add 228 calls and \$0.695073176; pose probes add 96 calls and \$0.162861212. The 192-call repeat/permutation study costs \$0.282100356. The full campaign totals 1,609 settled calls and \$3.097410574 under the original \$4.50 cap. No ledger is reset between studies. Local training and inference consume compute but do not incur API charges. These are historical campaign totals before the four-model paper expansion. The additional 156 requests and updated cumulative spend are generated from settled receipts in `expanded-evidence.json`; they are not included retroactively in the older campaign tables. The four new models cost \$0.523093880 across 156 settled requests. The cumulative campaign is 1,765 calls and \$3.620504454, below the unchanged \$4.50 cap. The new model allocation was capped at \$0.75; geometry audits and paper renders made no model calls.

8. Expanded Measured Model Evidence

The eight-model main matrix uses the identical 39 tasks from three already-observed development objects. Original GPT-4.1 mini, GPT-4.1, Gemini 2.5 Flash, and Qwen3-VL-32B responses are reused without rebilling. Claude Haiku 4.5, Gemini 3 Flash preview, Qwen3-VL-8B, and Llama 4 Maverick add 156 requests. Prompts and image bytes are hash-matched across rounds. New source observations are not claimed: these three objects already appear in the cumulative exposure register. Model endpoint IDs, catalog prices, provider routing, raw replies, latencies, tokens, and settled receipts are stored in `study/paper-models/`.

The same-source algorithm rows are generated from public inputs only. Copying applies to current-structure tasks; retrieval searches only training sources using public BOM or extent requirements without target-aware adaptation; symbolic rules solve relations/edits; greedy and height-sorted planners receive only the supplied target. Their supported-task scopes are explicit, and unsupported fields are not reported as failed attempts.

9. Finite-Grammar Ambiguity Contract

The new extension is versioned separately from v2. A fixed shell encloses a $2 \times W$ unknown layer tiled by identical red 1-by-2 bricks. All other layers and blue query poses are fixed. Widths 2, 4, and 6 and three stack heights yield nine parameterized families, not independently collected semantic objects. Every possible query pair is retained, not selected according to model outcomes. The two conditions share exactly the same query and family: the exterior condition withholds the internal layer, while support disclosure reveals its complete placement list.

`public.jsonl` supplies the grammar, known geometry, inventory, query IDs, and (only in the disclosed condition) support poses. It excludes expected labels, answer sets, and alternative hidden programs. Known geometry is symbolic, so this is a test of observation-consistent structural reasoning, not a claim of visual recognition. The public-input solver enumerates legal hidden tilings, forms all candidate contact graphs, and returns a definite answer only if every candidate agrees. It receives no evaluator target.

The interior enumeration uses first-empty-cell backtracking with horizontal and vertical domino placements. Independent recurrence counts validate enumeration completeness for the finite grammar. Every candidate passes both geometry implementations. Independent contact-edge intersections validate shared-support labels. The exact answer set is not claimed exhaustive over arbitrary brick programs or non-grid connectors.

False certainty counts definite yes/no on an undetermined input; unnecessary abstention counts undetermined on a determinate input. Malformed responses have zero ex-

act success and separate format failure. Neither conditional rate has a denominator on inapplicable cases. Because answer classes are unbalanced, overall accuracy, source-macro accuracy, and class-balanced accuracy must all be reported.

Pixel audit. All 60 layout states are rendered in four fixed views with the separate high-resolution paper renderer, for 240 frames. Thirty-six serve as the first-layout reference of their family/view, leaving 204 alternative comparisons. All alternative pixels match, with no tolerance. This does not certify equality under a different renderer, transparent parts, changed cameras, or actual photography. It also does not establish human agreement on the grammar or instructions.

Figure integrity. Thirty actual 3D renders form four new figure plates. Gallery examples are selected by median part count and deterministic group hash. Task examples preserve original geometry, colors, and evaluator IDs; all oracle targets are scored before rendering. Execution snapshots are verified legal prefixes, while the greedy dead end is checked against the original insertion rule. Fixed framing prevents camera rescaling from masquerading as a change in part pose. Exploded layers are explicitly display-only and are never substituted for ordinary model observations. Non-blank pixels, uncut framing, render recipes, and file hashes are recorded in `structure-evidence.json`.

Variant	Mini	GPT-4.1	Flash 2.5	Qwen 32	Haiku	Flash 3	Qwen 8	Llama 4
relations/contact	3/3	1/3	2/3	0/3	0/3	2/3	1/3	0/3
relations/separated	0/3	0/3	0/3	0/3	0/3	1/3	0/3	0/3
reconstruct/full	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3
generate/constraints	0/3	0/3	0/3	0/3	0/3	1/3	0/3	0/3
complete/suffix	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3
edit/rotate	0/3	0/3	0/3	0/3	0/3	1/3	0/3	0/3
edit/recolor	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3
edit/remove	3/3	3/3	3/3	3/3	3/3	3/3	3/3	2/3
plan/assemble	1/3	2/3	1/3	0/3	0/3	2/3	0/3	1/3
plan/disassemble	1/3	0/3	1/3	0/3	0/3	2/3	0/3	0/3
repair/none	3/3	1/3	2/3	1/3	2/3	0/3	1/3	0/3
repair/color	1/3	1/3	1/3	0/3	0/3	1/3	0/3	1/3
repair/shift	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3

Table 15. All measured task variants in the expanded screen. Each cell is successes out of the same three objects. No uncertainty interval or reliable ranking is inferred from three clusters.

Condition	Method	Correct / cases	Balanced acc.	Source-macro acc.
Exterior	Set-valued oracle	300/300	100.0	100.0
Exterior	Public-input enumeration	300/300	100.0	100.0
Exterior	Always no	210/300	50.0	57.8
Exterior	Always undetermined	90/300	50.0	42.2
Exterior	Single-witness answer	210/300	50.0	57.8
Disclosed	Set-valued oracle	300/300	100.0	100.0
Disclosed	Public-input enumeration	300/300	100.0	100.0
Disclosed	Always no	264/300	50.0	81.1
Disclosed	Always undetermined	0/300	0.0	0.0
Disclosed	Single-witness answer	300/300	100.0	100.0

Table 16. Measured algorithmic controls for the new ambiguity extension, percentages. Balanced accuracy averages only the answer classes present in that condition. The single-witness baseline commits to one valid layout even when the observation admits alternatives.

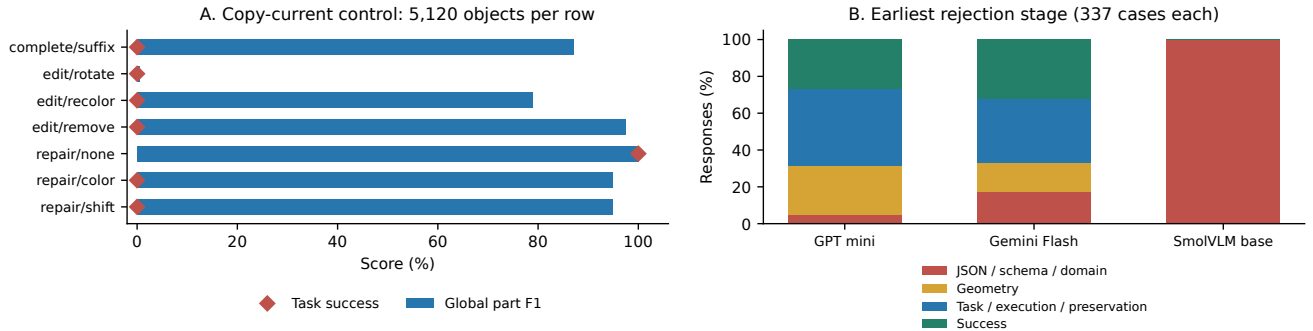


Figure 6. Previously measured no-operation and failure-stage diagnostics. The left panel uses the full casebank; the right uses the 337-case two-model/local-base study, not the new three-source matrix.

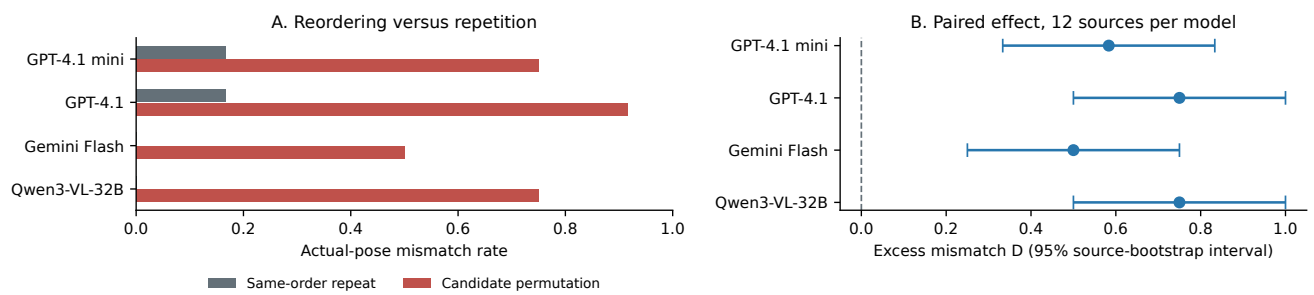


Figure 7. Measured same-order repetition versus candidate permutation on 12 shared development sources. Intervals resample source groups and do not constitute independent population confirmation.

10. Prospective Experiments: No Results Yet

This section is a work plan. Every result cell below is deliberately blank, and its machine-readable value in `experiments.json` is null. Tables are generated with an assertion that no prospective cell contains a measured number. They are not an extension of the completed pilots, a preregistration, or evidence of dataset acquisition. Tasks and models need final revision hashes and source manifests before execution. A model name here indicates an intended comparison, not confirmation that a future provider will serve an identical checkpoint.

10.1. Independent Main Evaluation

Table 17 gives the independent-confirmation layout. All methods must share source groups for paired comparison. Report ordinary RGB+BOM reconstruction/completion/repair separately from symbolic relations/editing/planning and constraint generation. Catalog recognition uses a separate type-cluster denominator. Each eventual score requires actual sources, failures, estimate, and 95% source-level interval. No pooled eight-task rank is prespecified. Sample size follows a predeclared precision target. For illustration, Hoeffding’s bound with a union bound over four means of $D_g \in [-1, 1]$ gives $n \geq \lceil 2 \log(8/\alpha)/\epsilon^2 \rceil$ independent sources for simultaneous absolute error at most ϵ with probability at least $1 - \alpha$. At $\alpha = .05, \epsilon = .10$, this is 1,016 sources and 16,256 calls for four models with four arms each. This is a conservative planning bound, not an acquired dataset, a guaranteed future effect, or a universal required sample size. Stratified/variance-based designs require their own frozen assumptions.

Method / condition	N	Part	Rel.	Recon.	Gen.	Compl.	Edit	Plan	Repair
GPT-4.1 mini									
GPT-4.1									
Gemini 2.5 Flash									
Qwen3-VL-32B									
Claude Haiku 4.5									
Gemini 3 Flash (preview)									
Qwen3-VL-8B									
Llama 4 Maverick									
Additional open VLM									

Table 17. **Unmeasured.** Independent eight-task confirmation. Blank cells are unrun results, not zero scores. N is the actual source denominator, with recognition reported separately by catalog type. Each cell requires an estimate and source-level interval; task variants and input conditions must also be broken out in the result artifacts.

10.2. Information and Representation

Table 18 is repeated per model on the same sources. Report the ordinary surface criterion for all arms, but full hidden reference success only with its information condition explicitly stated. Local pose narrows the requested output; candidates additionally restrict the answer space. They cannot isolate output length without that confound. For feedback versus reflection, reuse the same first response and charge the additional request once. Feedback remains target-blind; matching call counts alone does not match compute.

Method / condition	N	Success	Part F1	Surface F1	Format	Tokens
RGB+BOM / full structure						
BOM only / full structure						
Layer disclosure / structure						
Symbolic reference / structure						
RGB+BOM / local pose						
RGB+BOM / relative program						
RGB+BOM / legality feedback						
RGB+BOM / reflection only						

Table 18. **Unmeasured.** Paired information/representation ablations, separately per model. Cells need measured values and applicable denominators; blank is not zero.

10.3. Candidate Robustness and Debiasing

Table 19 compares original, repeated, and permuted actual pose selections on independent sources. Correct label balance and request ordering must be frozen. Missing responses reduce paired coverage, not the denominator of service availability. Report complete-pair effects alongside worst-case bounds: with N frozen sources, n complete pairs, and observed effect sum s , the all-source effect lies in $[(s - (N - n))/N, (s + (N - n))/N]$. This is a missing-outcome bound, not a confidence interval. For uniform independent random K -way choice, accuracy is $1/K$; repeat and permutation mismatch expectations are both $1 - 1/K$, so expected excess mismatch is zero. These are analytic controls, not empirical model results.

An option-debiasing baseline is motivated by existing selection-bias work and must be fitted on development data only. API conditions that do not expose token probabilities require a declared compatible alternative; one must not silently substitute a different method. Simulator-aware matching receives renderer access and should be reported with that additional resource.

Method / condition	Pairs	Accuracy	Repeat	Permute	D	95% CI
GPT-4.1 mini						
GPT-4.1						
Gemini 2.5 Flash						
Qwen3-VL-32B						
Dev-fitted option debiasing						
Known-renderer matching						

Table 19. **Unmeasured.** Independent candidate robustness. Repeat and Permute are pose-mismatch rates relative to the original, not error rates against ground truth; D is their paired difference.

10.4. Difficulty and Nuisance Factors

Table 20 must be instantiated per task/model. Cross piece count with policy, visibility, and graph complexity; do not call a marginal size trend causal. Nested prefixes stay in one source cluster. Render perturbations preserve geometry and receive new render/version hashes. Occlusion must be operationalized before selection (for example, visible cells or human-rated visibility), not assigned after observing model errors. Domain-specific local metrics include added-part F1, faulty-ID F1, or per-axis pose accuracy.

Method / condition	Sources	Success	Local F1	Format	95% CI
2–8 pieces / calibrated					
9–16 pieces / matched policy					
17–28 pieces / matched policy					
29–47 pieces / matched policy					
Canonical camera and palette					
Camera perturbation					
Palette perturbation					
Occlusion-matched structures					

Table 20. **Unmeasured.** Difficulty and camera/palette controls. Missing policy-size strata remain missing; they are not generated after seeing a desirable model ranking.

10.5. External Sources and Specialized Methods

Table 21 is contingent on actual licensed assets. Record source revision, file hashes, family IDs, permitted uses, unsupported part rejection, coordinate and connector mapping, and redistribution scope. A code license or a public paper does not grant rights to a separate structure collection. Do not flatten unsupported parts into cuboids and claim full-domain coverage.

MEPNet requires stepwise manual and component inputs, BrickGPT a compatible text-generation target, and BrickNet connector programs. Scores are meaningful only after task and input adaptation is audited. Native metrics and BrickAtlas metrics must be separately named. Unsupported cells should eventually be marked not applicable, not zero. No published score from another dataset may populate this table.

Method / condition	Sources	Types	Import rej.	Success	95% CI
Licensed human designs / VLM					
Licensed human designs / retrieval					
Manual-step subset / MEPNet adapter					
Text-generation subset / BrickGPT					
Connector subset / BrickNet					

Table 21. **Unmeasured.** External-domain and specialized methods. Import rej. is the rejected-source proportion with explicit reasons. No external structure collection or adapter run is claimed complete.

10.6. Human Metric Calibration

Table 22 requires at least two real independent, blinded reviewers and documented third-person adjudication when needed. Reviewers inspect public observations, candidate output, and the reference without automatic scores. An alternative answer may be correct despite differing from the reference. Preserve uncertain judgments and all original disagreements.

False acceptance uses human-rejected outputs as denominator; false rejection uses human-accepted outputs. Report both counts, source-cluster intervals, agreement before adjudication, unresolved coverage, and task strata. Missing denominators remain unestimable, not zero. Software verifies packet/version hashes, not real reviewer identity or independence. Current review packets cover 96 candidates from 48 cases and eight sources, with zero submitted labels; extending the packet is not equivalent to conducting the review.

Method / condition	Sources	Outputs	False acc.	False rej.	Agreement
Reference-correct outputs					
Alternative hidden layouts					
Near-correct geometry					
No-op edits and completions					
Normal and faulty repairs					
Executable and blocked plans					
Licensed external structures					

Table 22. **Unmeasured.** Human calibration categories. Outputs and independent sources are different denominators. No simulated or model-authored labels count as human evidence.

10.7. Effective Training and Replication

Table 23 describes a new study, not missing measurements from the ten completed jobs. First establish that the chosen checkpoint can use the interface and achieve nontrivial task performance. Freeze source-disjoint training/validation/test sets, model revision, supervised answer tokens, optimizer steps, and seeds. Also report image/input tokens, memory, and elapsed compute. Leave-task-out transfer needs an actual held-out-task improvement; lower validation loss alone is insufficient.

Independent reproduction uses a second executor and separately installed dependencies. Record commit, OS/architecture, versions, commands, exit codes, output hashes, and discrepancies. Separate offline rescoring, re-rendering, retraining, and new API replication: none certifies the others. Existing cumulative paid-call budgets remain in force.

Method / condition	Seeds	Success	Held-task gain	95% CI	Compute
Effective base checkpoint					
Single-task adaptation					
Multi-task adaptation					
Leave-edit-out adaptation					
Second-environment replication					

Table 23. **Unmeasured.** Effective training and independent replication. Held-task gain must identify its reference condition and paired source population.

10.8. Equal-budget tools and correction

Freeze one shared first response for each case. Distinguish a new model request from a replayed response. Compare equal call budgets and report actual token/time/renderer budgets. Only target-blind feedback is allowed; do not supply the hidden reference as tool feedback.

Method / condition	N	Success	Local F1	Legal prefix	Cost
One-shot / no tool					
One extra reflection					
One legality-check response					
Render own prediction + revise					
Active additional view					
Execution feedback + replan					
Fixed three-call tool budget					

Table 24. **Unmeasured.** Equal-budget tools and correction. Blank cells are unrun results, not zeros.

10.9. Representation and decoder factorial

Use identical source objects and observation payloads. Cross output representation with decoder support rather than changing them simultaneously. Relative offsets in global axes differ from local connector coordinates. A graph-backed adapter requires verified connector semantics; unsupported parts must be reported.

Method / condition	N	Format	Pose F1	Edge F1	Success
Absolute JSON / free decoding					
Absolute JSON / schema decoding					
Global-relative / free decoding					
Global-relative / schema decoding					
Connector graph / audited adapter					
Incremental placement / same images					

Table 25. **Unmeasured.** Representation and decoder factorial. Blank cells are unrun results, not zeros.

10.10. Ambiguity-aware model evaluation

New finite-grammar extension only, not v2. Evaluate exterior and support-disclosed arms separately with matched source families. Report class-balanced success because no/yes/undetermined are unequal. False certainty is conditional on ambiguous inputs; unnecessary abstention is conditional on determinate inputs. Do not equate nine grammar families with independent semantic sources.

Method / condition	Families	Balanced acc.	False certainty	Abstention	Disclosure gain
GPT-4.1 mini					
GPT-4.1					
Gemini 2.5 Flash					
Gemini 3 Flash (preview)					
Claude Haiku 4.5					
Qwen3-VL-8B / 32B					
Llama 4 Maverick					
Explicit candidate-layout enumeration					

Table 26. **Unmeasured.** Ambiguity-aware model evaluation. Blank cells are unrun results, not zeros.

10.11. Planning scope and harder action domains

Current vertical full-target planning is solved by height sorting; keep that as a ceiling control. New action domains need separately versioned geometry and independently verified witnesses. Do not add an obstacle or nonvertical action to the old evaluator and compare scores as if the task was unchanged.

Method / condition	Sources	Actions	Success	Search nodes	Time
Vertical full-target / neural baseline					
Partial-target / dependency inference					
Removal for access / repair sequence					
Workspace obstacles / search					
Typed connector / articulated action					

Table 27. **Unmeasured.** Planning scope and harder action domains. Blank cells are unrun results, not zeros.

10.12. Task-adapted comparisons and cross-benchmark validity

These are unimplemented or unrun comparisons, not published method scores. Obtain actual licensed assets and pin native evaluator revisions. Report native metrics and BrickAtlas metrics separately, with supported input/task mappings, rejection rates, and training overlap. Do not produce a pooled rank across incompatible benchmarks.

Method / condition	Sources	Native metric	Adapted success	Reject rate	Compute
BC-Bench / Brick-Composer					
MEPNet / manual-step adapter					
Neural Assembler / multi-view					
TreeSBA / multi-view adapter					
BrECS / budgeted completion					
BrickGPT / semantic generation					
BrickNet / connector generation					
SpatialBabel / scene-code control					

Table 28. **Unmeasured.** Task-adapted comparisons and cross-benchmark validity. Blank cells are unrun results, not zeros.

11. Artifact and Figure Reproduction

From the repository root, run the following with Node 22 and the documented Python figure dependencies:

```
node node_modules/tsx/dist/cli.mjs \
  benchmark/suite/study/paper-audit.ts
node benchmark/paper/generate-tables.mjs
node benchmark/paper/generate-study.mjs
node benchmark/paper/generate-audit.mjs
python benchmark/paper/generate-figures.py
node node_modules/tsx/dist/cli.mjs \
  benchmark/suite/study/ambiguity.ts --verify
node node_modules/tsx/dist/cli.mjs \
  benchmark/suite/study/paper-models.ts replay
node node_modules/tsx/dist/cli.mjs \
  benchmark/paper/generate-expanded.ts
node node_modules/tsx/dist/cli.mjs \
  benchmark/paper/render-structures.ts
python benchmark/paper/compose-structures.py
node node_modules/tsx/dist/cli.mjs \
  benchmark/suite/challenge/render.ts
python benchmark/suite/challenge/compose.py
node node_modules/tsx/dist/cli.mjs \
  benchmark/paper/generate-challenge.ts
```

Then compile both TeX documents using the instructions in `paper/README.md`. `audit-evidence.json` binds new tables to the census and prospective plan; `figure-evidence.json` binds plots to source files and output PDFs. The copy-control case-level file supports split/policy rescoring. The figure generator uses real archived PNGs, vector plots of observed data, and an explicitly disclosed cross-section of a verified witness.

The new audit does not modify frozen dataset generation, task labels, old responses, or legacy scorer behavior. It adds measurement coverage, not additional independent sources. Successful compilation, hash checks, and source-bound figures are technical release checks; they do not clear the independent scientific-validation gates. New three-dimensional plates use a separate renderer so historical evaluation images and their hashes remain unchanged. The optional render command needs the existing benchmark development server; all other commands above are offline after dependencies are installed.

12. Reference Manual and Evaluation Tracks

This appendix specifies the executable development benchmark, not a claim of independent validation. Dataset and model documentation follow the accountability motivation of Datasheets [9] and Model Cards [29]; accuracy, robustness and efficiency are distinct axes, as emphasized by HELM [22]. These references motivate reporting, not numerical guarantees.

12.1. Units and Version Boundaries

The frozen v2 population has 5,120 geometrically deduplicated assemblies and 119,070 placed part instances. Each source yields 13 task variants and 23 input-conditioned cases: $5120(2 + 3 + 1 + 3 + 3 + 2 + 9) + 150 = 117910$. The additional 150 queries are 25 known part types in six colors, not new assemblies. The 12 curated sources contain 193 parts and eight original demonstrations. The six challenge sources contain 329 parts and twelve task contracts. The audit gallery adds 24 illustrative plan/repair controls for the curated sources, and reuses the twelve revised challenge cases: 36 positive/negative control pairs on 18 sources. These controls are not paid or learned-model predictions. All 5,120 frozen source thumbnails are published on 80 offline catalog pages, each with two actual case IDs (ordinary reconstruction and assembly planning), totaling 10,240 case links. The full 117,910 tasks remain in the compressed shards.

Challenge revision 1 is retained under `challenge-cases/` for provenance and deprecated for new scoring. Revision 2 under `challenge-cases-v2/` changes input and scoring contracts, and receives new versioned case IDs. No historical model score is translated into a revision-2 result. The original v2 population is unchanged. Exact hash reproducibility is not reproducibility of stochastic API answers.

12.2. Task Contracts

Family	Public observation	Required response	Acceptance and negative control
Parts	Two isolated RGB views	Type, color, stud count	Exact fields; wrong type/color/count fails
Relations	Assembly JSON and queried IDs	Connected, above, contact studs, path length	Typed fields; side proximity is not a stud join
Reconstruction	RGB+BOM, layers, or symbolic reference	Full typed placement multiset	Declared surface or full match; shifted origin fails
Generation	Six numeric constraints	Any satisfying assembly	Legal, anchored, extent/count/color/graph constraints; not witness matching
Completion	Current assembly and reference	Completed full assembly	Target rule plus preserved input and additions; copy-current fails
Edit	Current assembly and operation	Full edited assembly	Requested change and no collateral change; no-operation fails
Plan	Full target and assemble/disassemble direction	Ordered IDs	Original order executes; missing, duplicate and blocked steps fail
Repair	Current assembly, BOM and reference	Full assembly and wrong current IDs	Restore, preserve, localize; empty fault set required in normal controls

Visual information. Ordinary v2 RGB uses four assembled views and a colored positive-axis first-hit surface criterion plus BOM and nominal legality. This proxy is not a theorem of pixel equivalence. Layer disclosure supplies every bottom-Y layer and requires full geometry; symbolic reference supplies full coordinates and is privileged. Revised challenge scene reconstruction, distributed completion and multi-fault repair now add layer disclosure, because their evaluator requires complete geometry. We have not established uniqueness from the old exterior images. The corrected claim is an input-sufficiency safeguard, not a measured ambiguous-pair discovery.

12.3. Advanced Contracts and Corrections

Task	Revision-2 contract	Failure tested
Recovery	Initial/target JSON, at most 256 remove/place actions; replay without reordering	Continue without removing decks: blocked corridor
Support counterfactual	Delete the public ID set simultaneously; iteratively remove unsupported bodies	Omit cascading losses
Compound edit	All blue pieces to red, two bench removals, two explicit additions	Partial instruction or missing addition
Multi-fault repair	Current structure, BOM, full layer observations; wrong current-ID set	Correct geometry with incomplete fault set
Inspection	Query, returned observation, final answer; success and query cost separate	Answer before query, invalid/repeated query
Graph reasoning	Full public geometry; shortest contact path and all articulation pieces	Wrong path, missing or duplicate articulation ID
Distributed completion	Eight missing columns across two tiers; disclosed layers	Copy-current or collateral change
Module transform	Public module IDs and translation; preserve lower tier	Omitted module piece or wrong translation
Step selection	Public current geometry and nine complete candidate poses	Any illegal selected candidate or omitted legal candidate
Pose	Selected post, current structure and reference; yaw modulo footprint symmetry	Wrong coordinate or yaw outside allowed domain
Scene reconstruction	53-part market; four assembled views plus complete layers	Missing piece or wrong pose
Constrained redesign	Public foundation, inventory, extent and count thresholds	Missing anchor, inventory overrun or unmet constraint

Inspection is not an optimality theorem. `InspectionEpisode.query()` accepts one known query, returns a renderer recipe or an explicitly privileged structure/graph, and logs cost. A trusted renderer must convert a recipe to PNG before sending it to a VLM. Submission before querying and repeated submissions are rejected. Layer and full-symbolic disclosure both satisfy the explicit disclosure policy, with costs 2 and 4. This is a small executable information-acquisition control, not a calibrated test of optimal active perception. No robot, network inference runner, or learned inspection policy is claimed. Tool use is related to ReAct [47]; single-response refinement controls are related to Self-Refine [26], but our receipts do not record private internal reasoning.

Graph and support meaning. Shortest paths use undirected stud contacts. Articulation points use depth-first low-link logic [39] and are independently checked by deleting each vertex and recounting components. Cascading support deletion is a monotone discrete rule: remove any elevated part lacking direct support, repeat to a fixed point. It is not gravity integration, force equilibrium, clutch strength or damage simulation. The redesign “roof” metric counts occupied three-dimensional cells of parts whose bottom $Y \geq 7$; it is a volume proxy and does not verify an architectural roof.

12.4. Metrics and Edge Cases

Let T and A be target and submitted part multisets; let m be the one-to-one match count by type, color and footprint pose, modulo per-part yaw symmetry. Arbitrary IDs are ignored for geometric matching, but queried and faulty current IDs remain task-specific handles. There is no silent global alignment.

$$P_{\text{part}} = m/|A|, \quad R_{\text{part}} = m/|T|, \quad F_{\text{part}} = 2m/(|A| + |T|), \quad (1)$$

$$\text{IoU} = |V(T) \cap V(A)|/|V(T) \cup V(A)|, \quad \text{Preserve} = |\{p \in P : k(p) \in k(A)\}|/|P|. \quad (2)$$

Here P is the protected input region. Set denominators equal to zero receive the implementation’s explicit empty-set convention, not a dropped observation: the pairwise F1 helper returns 1 when both sets are empty; a required full structure cannot

be empty and still succeed. Invalid schemas receive format=0 and success=0. All selected missing responses remain failures in applicable means.

Metric	Definition	Interpretation boundary
BOM F1	Multiset type/color overlap, pose ignored	High BOM does not prove reconstruction
Edge F1	Matched endpoint pair and contact-stud count	Nominal contacts only
Surface F1	Colored first-hit grid-cell set overlap	Not rendered image similarity
Addition F1 (v2)	Matched missing-target parts against submitted additions	Copy-current must not earn added-region credit
Changed-target recall (challenge)	Fraction of changed target pieces present	Not addition precision; deletions also require final exact
Preservation	Protected input pieces retained by typed pose	Extra/missing pieces still affect final success
Fault-set exact	Exact unique current-ID set	No-fault controls have an empty set
Legal prefix	Successfully executed actions / submitted actions	Always pair with finalExact and coverage; short valid prefix is not success
Pose field accuracy	Correct X,Y,Z and equivalent yaw / 4	Square yaw classes collapse; not robot SE(3) error
Constraint success	Conjunction of validity, extent, inventory, anchors, volume and color-count checks	Multiple witnesses accepted
Query cost	Cost of the executed query	Success is not forced to use the cheapest query

Global F1 is intentionally insufficient: copied v2 completion inputs reach 87.1% part F1 and copied removal inputs reach 97.4% without doing the task. These controls are motivated by shortcut-learning concerns [10]. Percentages from different tasks or information conditions are not averaged into one rank. Report cluster bootstrap intervals over source groups [8]; Wilson intervals [45] are descriptive only where binomial trial assumptions are justified. Derived cases and reruns of the same source are not independent.

12.5. Method-Specific Tracks, Shared Comparison Rules

Different output interfaces deserve different tracks, not easier private questions for a favored model. Within a track all systems receive the same case IDs, observation bytes, feedback allowance, output budget, and scorer revision.

Track	Applicable methods	Task category	What may be compared
Symbolic	LLM, graph solver, program synthesizer	Graph, counterfactual, compound edit, module, selection, redesign	Same public geometry or numeric constraints
Disclosed vision	VLM, reconstruction specialist	Reconstruction, completion, multi-fault repair	Same assembled and layer images; no hidden coordinates
Local pose	VLM or geometric estimator	Selected-piece placement	Same target views and current assembly
Open-loop recovery	LLM/search planner	Remove-and-rebuild programs	Same 256-action cap, initial state and goal
Inspection	Tool agent	Query followed by answer	Returned observation and cost logged; not merged with one-shot
Robot control	OpenVLA/RT-2-style policy	Continuous robot execution	Not implemented; no score or eligibility claim

OpenVLA [19] and RT-2 [6] predict robot actions in an embodiment, unlike digital integer-grid placements. Their appearance in a method plan does not mean an adapter or experiment exists. BrickNet’s LDraw representation [20, 21], BrickGPT’s physical generation [35], and BC-Bench’s manual-based selection/pose [23] likewise need audited adapters; unavailable methods are marked unsupported rather than assigned synthetic zero scores.

12.6. Model and Software Sources

The eight-model development matrix uses GPT-4.1 and mini [31], Gemini variants [11], Qwen3-VL checkpoints [36], Claude Haiku 4.5 [2], and Llama 4 Maverick [27]. Official model pages identify the families; exact runtime model IDs, providers, prices, input hashes and 2,200-token caps are in the archived protocols and OpenRouter receipts [32]. These living web pages are not evidence of fixed server weights or deterministic sampling. SmolVLM and Qwen text pilots retain their own citations and cohorts.

Three.js [41] renders the grid geometry. Training uses PyTorch [33], Transformers [46] and LoRA [14]; local inference uses MLX [3]. The reproducibility stack additionally uses Node/TypeScript [30], tsx, Vite [42], Playwright [28], pngjs, Python, Pillow [34], Matplotlib [17], NumPy [13], PyMuPDF [4], Tectonic [40] and the official author kit [7]. Their pinned versions are in lockfiles, `requirements.txt` and `BUILD.md`; no numerical performance property is inferred from those tools.

12.7. Efficiency, Robustness, Fairness and Contamination

For each future inference report separately: model latency, rendering latency, parsing/scoring latency, input/output tokens, query/action count, billed USD, hardware, numerical precision and peak local device memory. Remote model memory is unobservable and remains null. The protocol includes these fields without filling unrun measurements. Report per-track p50/p95 latency only after sufficient runs. Match total budgets when comparing feedback or search; matching optimization steps does not match input tokens or compute.

Robustness should cross repeated identical input, candidate permutation, ID renaming, equivalent yaw, camera/palette perturbation and no-image controls. Permutations preserve the answer mapping; missing images alter the information condition. Report consistency separately from correctness. The current paired order study is a development diagnostic, not proof of model-wide reliability.

Group splits exclude full geometric duplicates across train/test under the defined normalization. MinHash/LSH screening [5] does not establish exhaustive subassembly separation, nor exclusion from unknown pretraining corpora. All current cases and this gallery are public: they must not be called hidden tests. Future confirmation requires unseen source groups, frozen selection before outcomes, an exposure ledger, timestamped revisions and independent labels. No hypothetical dynamic update mechanism is counted as implemented contamination protection.

12.8. Audit Scope

Four revision-1 defects were independently confirmed by two code-review passes: missing candidate poses, an inspection task without returned observations, primitive field coercion, and incorrect square-yaw rejection. They are regression-tested in revision 2. Exterior-image uniqueness was not proven false for all examples; additional layers make the revised exact-match contract explicit. Technical validation, independent code checks, semantic human review and model performance are different evidence types. No real human judgments or new paid challenge evaluations were created by this revision.

12.9. Nine-Dimension Internal Readiness Assessment

This is an author-side, evidence-based assessment, not a conference review or an empirical model score. The 0–10 anchors are: absent (0), concept/demo (2), narrow-domain executable (4), repeated technical evidence (6), independent cross-domain evidence (8), and sustained external validation (10). No average is reported because a validity failure cannot be offset by good engineering.

Dimension	Before	After	Evidence and remaining gate
Representativeness	3	3	Rectangular grid, mainly architecture/furniture; licensed real workflows missing
Sensitivity	2	2	Measured floors/ceilings; no challenge-model runs to justify a gain
Multidimensional coverage	4	5	Explicit tracks, metrics, resource fields and controls; efficiency not yet fully measured
Difficulty gradient	3	4	All 8–63-part authored objects inspectable; no human difficulty calibration
Reproducibility	6	6	Hash replay and reproducible figures; independent installation and API repeats outstanding
Fairness	4	6	Input sufficiency, strict types and yaw corrections; specialist/provider confounds remain
Transparency	7	8	Full source gallery, task links, positive/negative scores and recorded limitations
Contamination resistance	2	2	Public development data; unknown pretraining exposure, no hidden confirmation set
Version stability	5	7	Deprecated v1 preserved, v2 IDs and contracts explicit, regression tests added

These scores prioritize the next scientific work: frozen unseen-source comparisons with effective baselines, licensed external designs, blinded real human calibration, and independent reproduction. Documentation alone does not raise representativeness, model sensitivity or contamination resistance.

13. All Authored Models: Scored Case Studies

These are constructed positive and negative evaluator controls, not model predictions. The 18 source models remain 18 sources; their 36 demonstration cases are not independent samples. All 5,120 frozen procedural sources are shown separately in the 80-page offline gallery.

13.1. Garden Bench

Source: `easy-garden-bench`. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

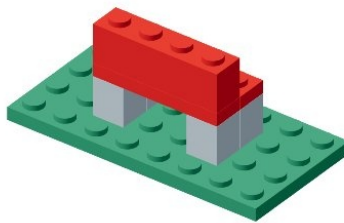
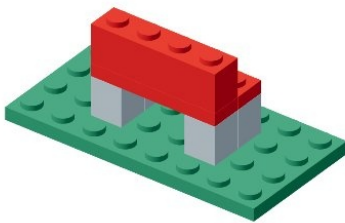
plan. Case `152306ac476e4708d75f7668`.

Given the full target, return all IDs in an executable vertical insertion order. The negative reverses the order.

Input / context

Positive reference

Constructed negative



Strict success: positive 1, negative 0. Rejection: `unsupported`, `incomplete`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-01.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

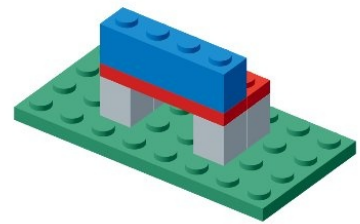
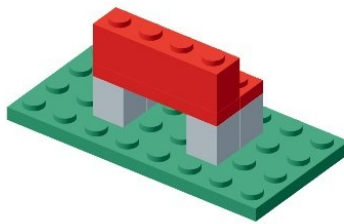
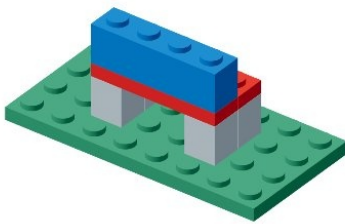
repair. Case `3a48a42a7c4f4265c5bc9d3a`.

Given current and full symbolic reference, restore a wrong color and report its current ID. The negative copies current.

Input / context

Positive reference

Constructed negative



Strict success: positive 1, negative 0. Rejection: `target_mismatch`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-02.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

13.2. Four-Step Staircase

Source: `easy-staircase`. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

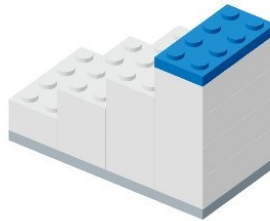
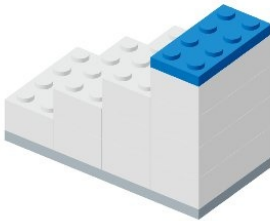
plan. Case `f6d7ad54390ef8403fa59152`.

Given the full target, return all IDs in an executable vertical insertion order. The negative reverses the order.

Input / context

Positive reference

Constructed negative



Strict success: positive 1, negative 0. Rejection: `unsupported`, `incomplete`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-03.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

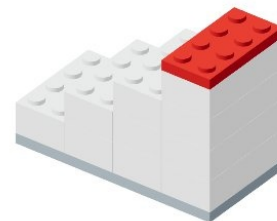
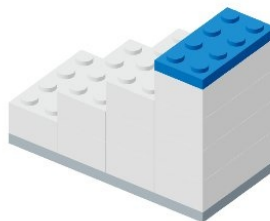
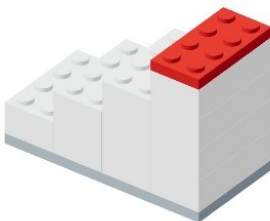
repair. Case `cf27b03347e69a62e88adb00`.

Given current and full symbolic reference, restore a wrong color and report its current ID. The negative copies current.

Input / context

Positive reference

Constructed negative



Strict success: positive 1, negative 0. Rejection: `target_mismatch`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-04.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

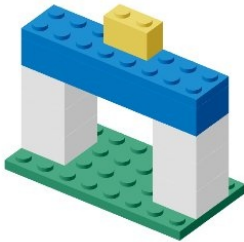
13.3. Garden Gate

Source: `easy-garden-gate`. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

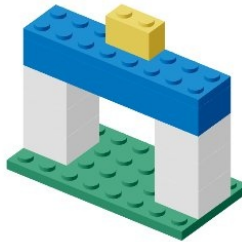
plan. Case `3ba7340d88ba53b511995760`.

Given the full target, return all IDs in an executable vertical insertion order. The negative reverses the order.

Input / context



Positive reference



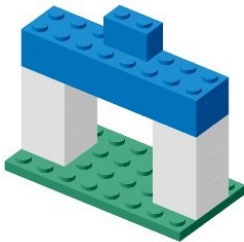
Constructed negative

Strict success: positive 1, negative 0. Rejection: `unsupported`, `incomplete`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-05.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

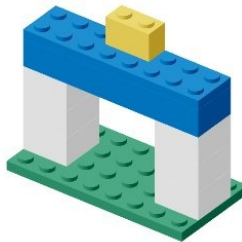
repair. Case `f1e7a66185b2390ffaec600f`.

Given current and full symbolic reference, restore a wrong color and report its current ID. The negative copies current.

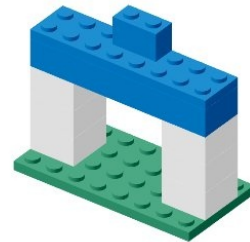
Input / context



Positive reference



Constructed negative



Strict success: positive 1, negative 0. Rejection: `target_mismatch`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-06.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

13.4. Picnic Table

Source: `easy-picnic-table`. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

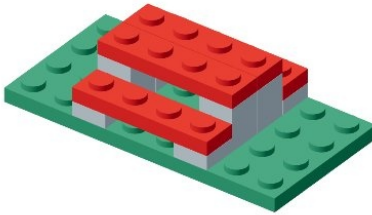
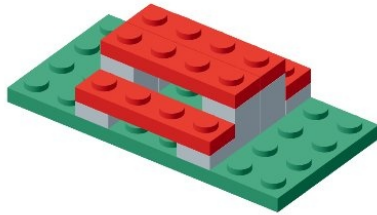
plan. Case `c9ef6834c5de4d543be5b056`.

Given the full target, return all IDs in an executable vertical insertion order. The negative reverses the order.

Input / context

Positive reference

Constructed negative



Strict success: positive 1, negative 0. Rejection: `unsupported`, `incomplete`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-07.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

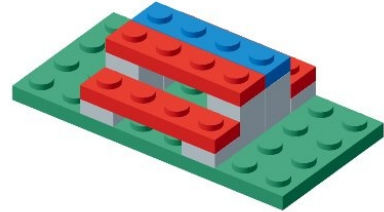
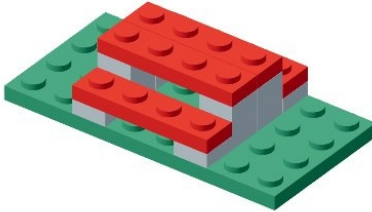
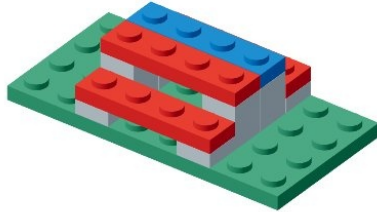
repair. Case `92deb3d852a09a10aac8b5c4`.

Given current and full symbolic reference, restore a wrong color and report its current ID. The negative copies current.

Input / context

Positive reference

Constructed negative



Strict success: positive 1, negative 0. Rejection: `target_mismatch`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-08.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

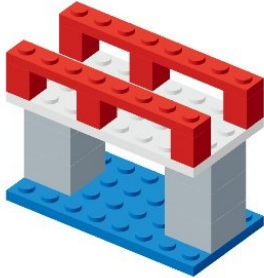
13.5. Canal Bridge

Source: medium-canal-bridge. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

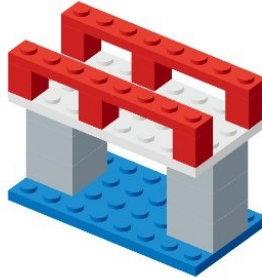
plan. Case 5c96ed823332734ad4af5053.

Given the full target, return all IDs in an executable vertical insertion order. The negative reverses the order.

Input / context



Positive reference



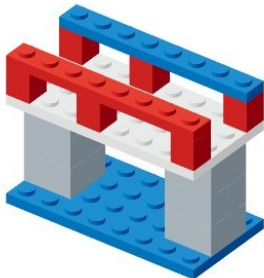
Constructed negative

Strict success: positive 1, negative 0. Rejection: unsupported, incomplete. Machine-readable inputs, both answers and all metrics are in review-gallery/case-09.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

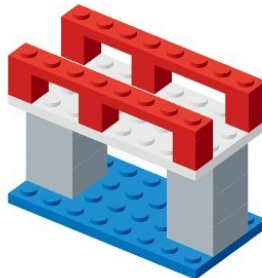
repair. Case 62c0448ef5755ddb2d3e7ad4.

Given current and full symbolic reference, restore a wrong color and report its current ID. The negative copies current.

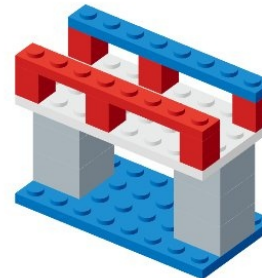
Input / context



Positive reference



Constructed negative



Strict success: positive 1, negative 0. Rejection: target_mismatch. Machine-readable inputs, both answers and all metrics are in review-gallery/case-10.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

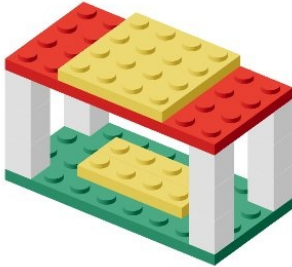
13.6. Park Pavilion

Source: medium-pavilion. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

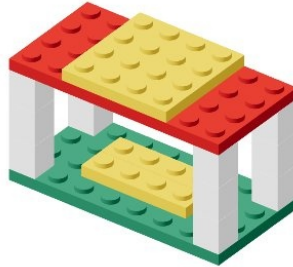
plan. Case 025e602d17a4226de90cdd8e.

Given the full target, return all IDs in an executable vertical insertion order. The negative reverses the order.

Input / context



Positive reference



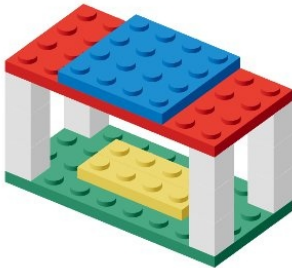
Constructed negative

Strict success: positive 1, negative 0. Rejection: unsupported, incomplete. Machine-readable inputs, both answers and all metrics are in review-gallery/case-11.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

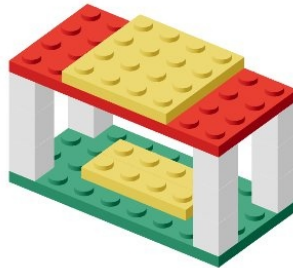
repair. Case ff174846fe2b77b68cafa0b1.

Given current and full symbolic reference, restore a wrong color and report its current ID. The negative copies current.

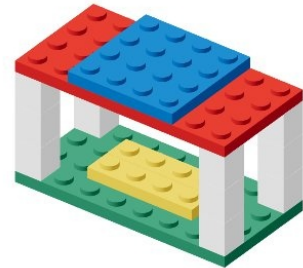
Input / context



Positive reference



Constructed negative



Strict success: positive 1, negative 0. Rejection: target_mismatch. Machine-readable inputs, both answers and all metrics are in review-gallery/case-12.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

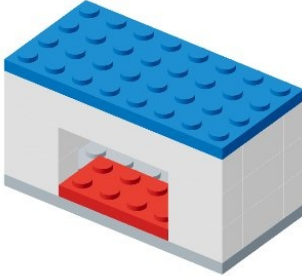
13.7. Townhouse

Source: medium-townhouse. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

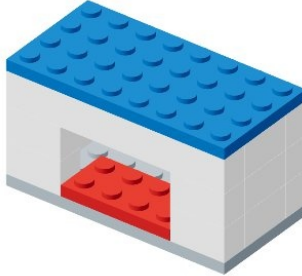
plan. Case 39fc27ace5ece93b349ee3a8.

Given the full target, return all IDs in an executable vertical insertion order. The negative reverses the order.

Input / context



Positive reference



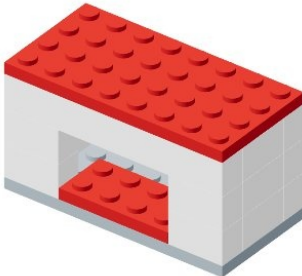
Constructed negative

Strict success: positive 1, negative 0. Rejection: unsupported, incomplete. Machine-readable inputs, both answers and all metrics are in review-gallery/case-13.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

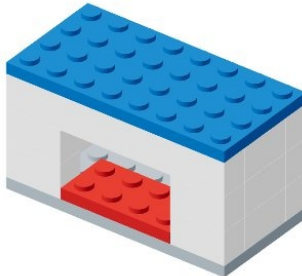
repair. Case 04131146e62411747ad0ebc5.

Given current and full symbolic reference, restore a wrong color and report its current ID. The negative copies current.

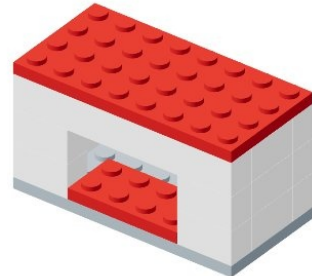
Input / context



Positive reference



Constructed negative



Strict success: positive 1, negative 0. Rejection: target_mismatch. Machine-readable inputs, both answers and all metrics are in review-gallery/case-14.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

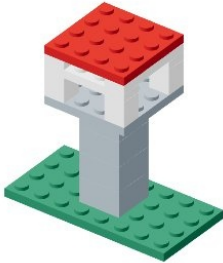
13.8. Watchtower

Source: medium-watchtower. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

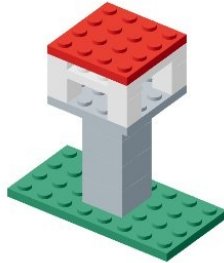
plan. Case 403b52f444583f361b1d37f3.

Given the full target, return all IDs in an executable vertical insertion order. The negative reverses the order.

Input / context



Positive reference



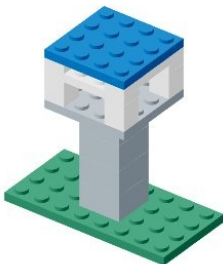
Constructed negative

Strict success: positive 1, negative 0. Rejection: unsupported, incomplete. Machine-readable inputs, both answers and all metrics are in review-gallery/case-15.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

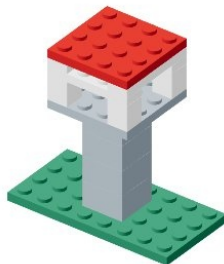
repair. Case 6d49bea3c782adb17f8760c0.

Given current and full symbolic reference, restore a wrong color and report its current ID. The negative copies current.

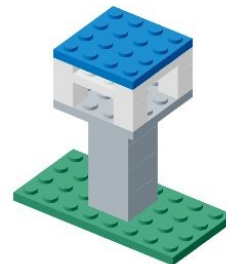
Input / context



Positive reference



Constructed negative



Strict success: positive 1, negative 0. Rejection: target_mismatch. Machine-readable inputs, both answers and all metrics are in review-gallery/case-16.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

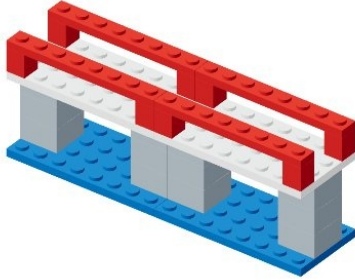
13.9. Double-Span Bridge

Source: `hard-double-span-bridge`. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

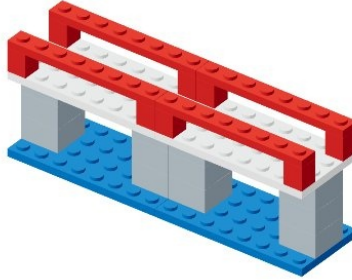
plan. Case `dfacbf9787f98c34c8203850`.

Given the full target, return all IDs in an executable vertical insertion order. The negative reverses the order.

Input / context



Positive reference



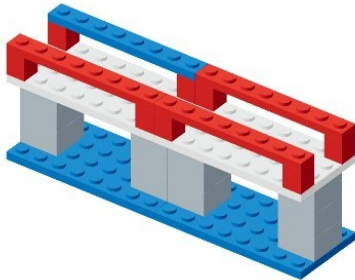
Constructed negative

Strict success: positive 1, negative 0. Rejection: `unsupported`, `incomplete`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-17.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

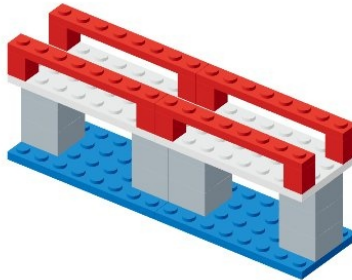
repair. Case `bb95cd1c7931fd1e15402b73`.

Given current and full symbolic reference, restore a wrong color and report its current ID. The negative copies current.

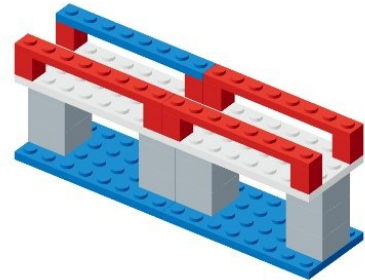
Input / context



Positive reference



Constructed negative



Strict success: positive 1, negative 0. Rejection: `target_mismatch`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-18.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

13.10. Lighthouse

Source: hard-lighthouse. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

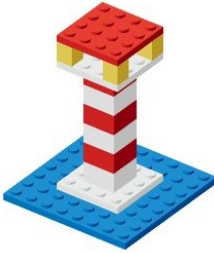
plan. Case 2ae74752a90a40fcc98cbb78.

Given the full target, return all IDs in an executable vertical insertion order. The negative reverses the order.

Input / context

Positive reference

Constructed negative



Strict success: positive 1, negative 0. Rejection: unsupported, incomplete. Machine-readable inputs, both answers and all metrics are in review-gallery/case-19.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

repair. Case 52cdc0d3b4abf89218b9254e.

Given current and full symbolic reference, restore a wrong color and report its current ID. The negative copies current.

Input / context

Positive reference

Constructed negative



Strict success: positive 1, negative 0. Rejection: target_mismatch. Machine-readable inputs, both answers and all metrics are in review-gallery/case-20.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

13.11. Railway Station

Source: `hard-railway-station`. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

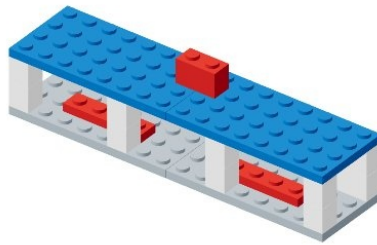
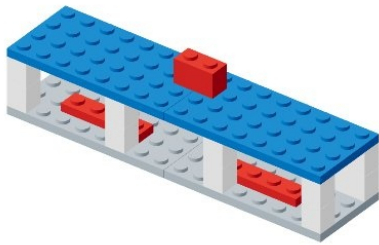
plan. Case `4cbafe8643409116636b5a60`.

Given the full target, return all IDs in an executable vertical insertion order. The negative reverses the order.

Input / context

Positive reference

Constructed negative



Strict success: positive 1, negative 0. Rejection: `unsupported`, `incomplete`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-21.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

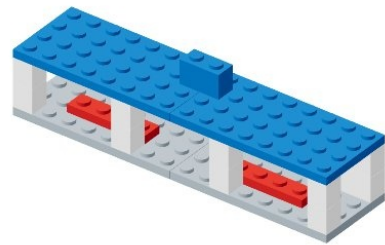
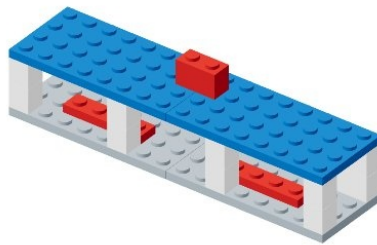
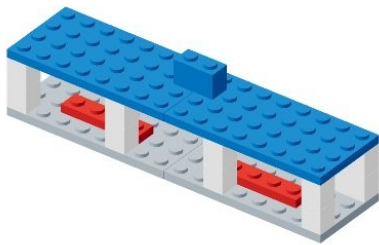
repair. Case `b45983f48340bdb36efb4625`.

Given current and full symbolic reference, restore a wrong color and report its current ID. The negative copies current.

Input / context

Positive reference

Constructed negative



Strict success: positive 1, negative 0. Rejection: `target_mismatch`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-22.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

13.12. Two-Tier Pagoda

Source: `hard-two-tier-pagoda`. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

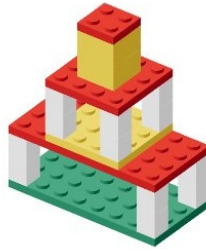
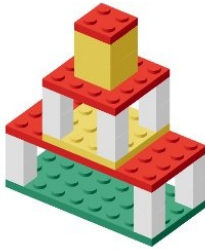
plan. Case `de20e6915476e83efe0342d7`.

Given the full target, return all IDs in an executable vertical insertion order. The negative reverses the order.

Input / context

Positive reference

Constructed negative



Strict success: positive 1, negative 0. Rejection: `unsupported`, `incomplete`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-23.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

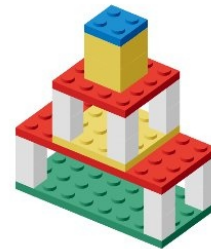
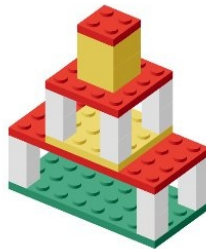
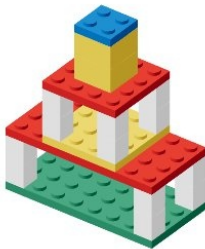
repair. Case `9e1b095a051a63df43d8d852`.

Given current and full symbolic reference, restore a wrong color and report its current ID. The negative copies current.

Input / context

Positive reference

Constructed negative



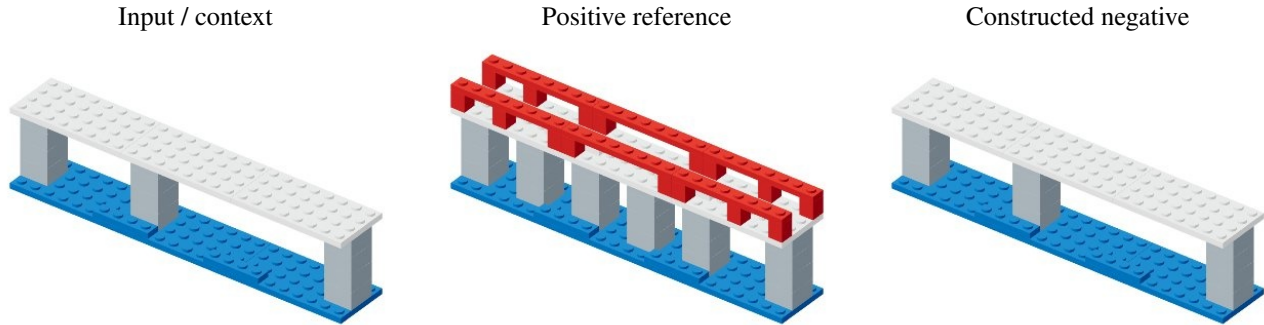
Strict success: positive 1, negative 0. Rejection: `target_mismatch`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-24.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

13.13. Three-Span Truss Bridge

Source: `expert-truss-bridge`. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

recovery-plan. Case 4525bbe39009b121d8181a01.

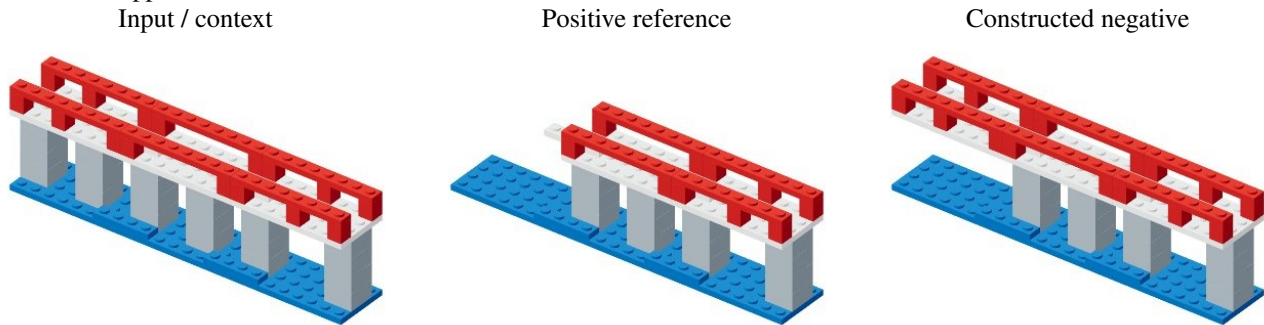
Recover from a legal but blocked deck-first state. Return remove/place actions that reach the complete target without reordering by the evaluator.



Strict success: positive 1, negative 0. Rejection: `illegal_action`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-25.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

support-counterfactual. Case 03102cc267961f8f67a6c1fd.

If the declared two support columns disappear simultaneously, report every additional piece removed by iterative loss of direct stud support.



Strict success: positive 1, negative 0. Rejection: `set_mismatch`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-26.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

```
Positive: {"unsupportedIds": ["deck-1", "rail-1", "rail-4", "rail-post-1", "rail-post-10",  
"rail-post-11", "rail-post-2", "rail-post-3", "rail-post-9"]}  
Negative: {"unsupportedIds": []}
```

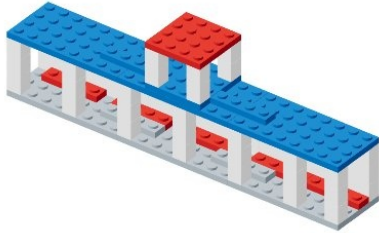
13.14. Grand Railway Terminal

Source: `expert-grand-terminal`. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

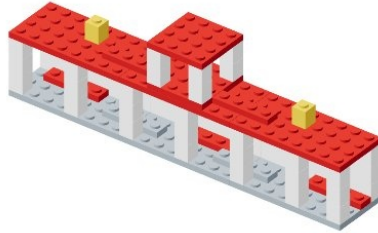
compound-edit. Case `784592203ea29dc17078f9eb`.

Apply all clauses atomically: recolor the blue canopy red, remove bench-2 and bench-4, add the two specified yellow roof lights, and preserve every other part.

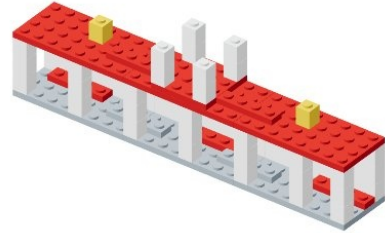
Input / context



Positive reference



Constructed negative

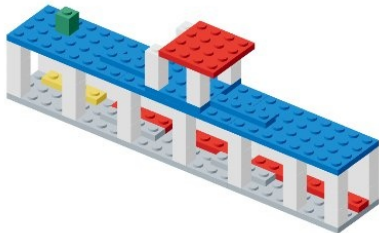


Strict success: positive 1, negative 0. Rejection: `target_mismatch`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-27.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

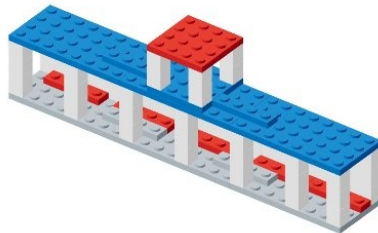
multi-fault-repair. Case `e8c2121ecc177d0237e011dd`.

Repair three simultaneous faults: one recolor, one shifted part, and one extra part. Return the corrected structure and all wrong CURRENT IDs. Layer disclosure is supplied: recover complete geometry up to piece yaw symmetry, not exterior RGB alone.

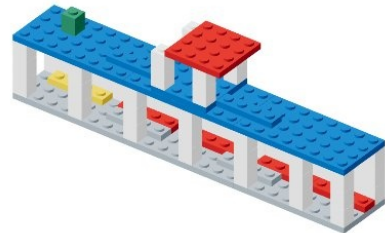
Input / context



Positive reference



Constructed negative



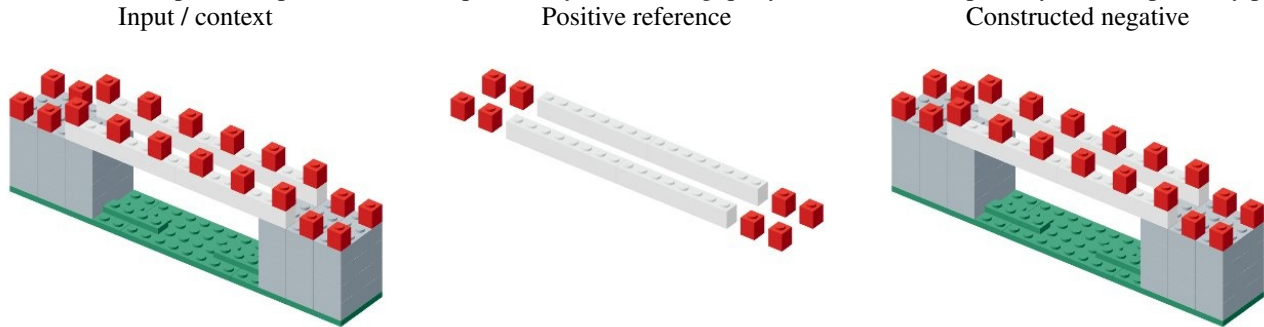
Strict success: positive 1, negative 0. Rejection: `target_mismatch`, `fault_mismatch`. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-28.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

13.15. Fortress Gatehouse

Source: expert-fortress-gatehouse. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

active-inspection. Case 6f5b964a86860cd2e99722de.

Select a query, read its returned observation, then report how many parallel rows the white gate beams occupy. Each row can contain multiple beam parts. Success requires a layer-disclosing query; cost is scored separately, not an optimality proof.

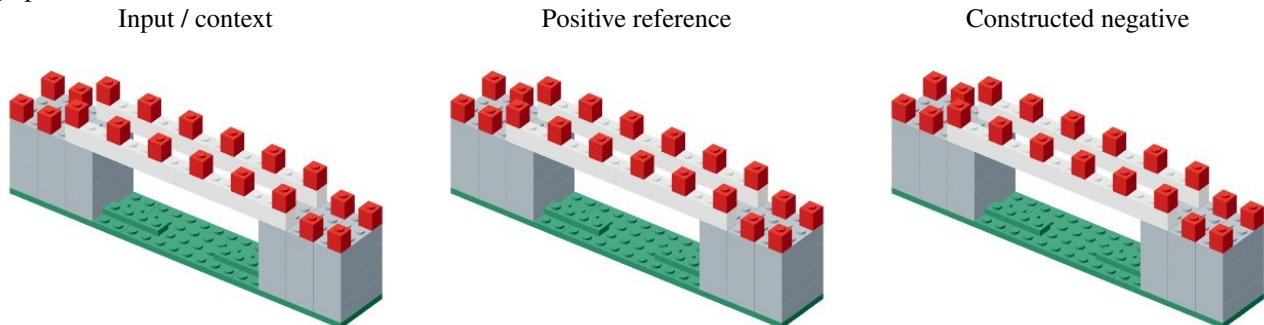


Strict success: positive 1, negative 0. Rejection: answer_mismatch. Machine-readable inputs, both answers and all metrics are in review-gallery/case-29.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

```
Positive: {"queryId": "layer-y-13", "answer": "two-parallel-beams"}
Negative: {"queryId": "unknown", "answer": "two-parallel-beams"}
```

graph-reasoning. Case 2057760ea52daf7c9b82ccb3.

Return the shortest stud-contact path length between the queried merlons and every articulation piece in the full contact graph.



Strict success: positive 1, negative 0. Rejection: answer_mismatch. Machine-readable inputs, both answers and all metrics are in review-gallery/case-30.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

```
Positive: {"shortestPath": 14, "articulationIds": ["base-1", "base-2", "base-3", "base-seam-1", "base-seam-2", "gate-beam-1", "gate-beam-2", "gate-beam-3", "gate-beam-4", "tower-1", "tower-10", "tower-11", "tower-12", "tower-13", "tower-14", "tower-15", "tower-16", "tower-17", "tower-18", "tower-19", "tower-2", "tower-20", "tower-21", "tower-22", "tower-23", "tower-24", "tower-3", "tower-4", "tower-5", "tower-6", "tower-7", "tower-8", "tower-9"]}
Negative: {"shortestPath": 15, "articulationIds": ["base-1", "base-2", "base-3", "base-seam-1", "base-seam-2", "gate-beam-1", "gate-beam-2", "gate-beam-3", "gate-beam-4", "tower-1", "tower-10", "tower-11", "tower-12", "tower-13", "tower-14", "tower-15", "tower-16", "tower-17", "tower-18", "tower-19", "tower-2", "tower-20", "tower-21", "tower-22", "tower-23", "tower-24", "tower-3", "tower-4", "tower-5", "tower-6", "tower-7", "tower-8", "tower-9"]}
```

13.16. Stepped Temple Complex

Source: advanced-stepped-temple. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

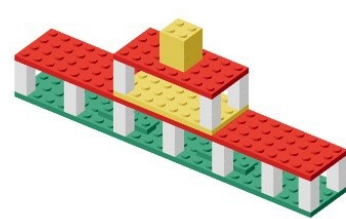
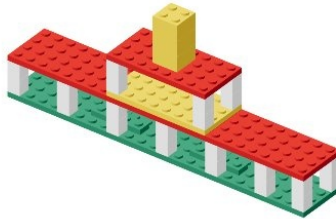
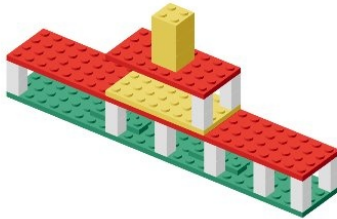
distributed-completion. Case 0482601199de9b79b7208a9d.

Restore eight missing columns distributed across both tiers. Preserve every supplied part exactly; the missing region is not a suffix. Layer disclosure is supplied: recover complete geometry up to piece yaw symmetry, not exterior RGB alone.

Input / context

Positive reference

Constructed negative



Strict success: positive 1, negative 0. Rejection: target_mismatch. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-31.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

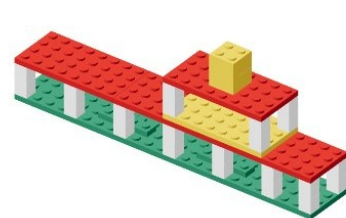
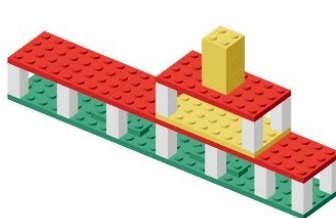
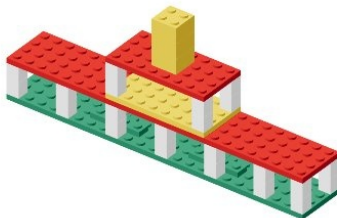
module-transplant. Case 1ed677903af5f888740674f9.

Move the complete upper pavilion four studs toward +X as a rigid module. Preserve its internal layout and every lower-tier part.

Input / context

Positive reference

Constructed negative



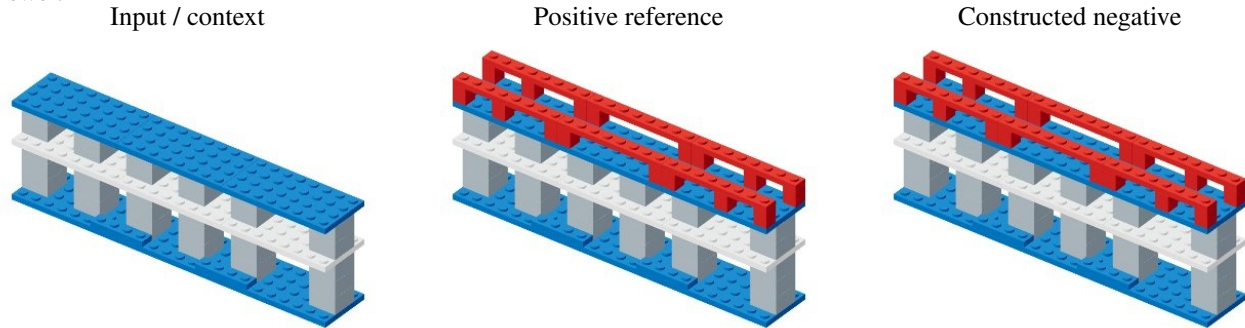
Strict success: positive 1, negative 0. Rejection: target_mismatch. Machine-readable inputs, both answers and all metrics are in `review-gallery/case-32.json`. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

13.17. Double-Deck Aqueduct

Source: expert-double-deck-aqueduct. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

step-selection. Case c5ba6f85a9a06e3162407ac6.

Select every candidate that can legally be inserted next into the current partial aqueduct. Do not use the target order as an answer.

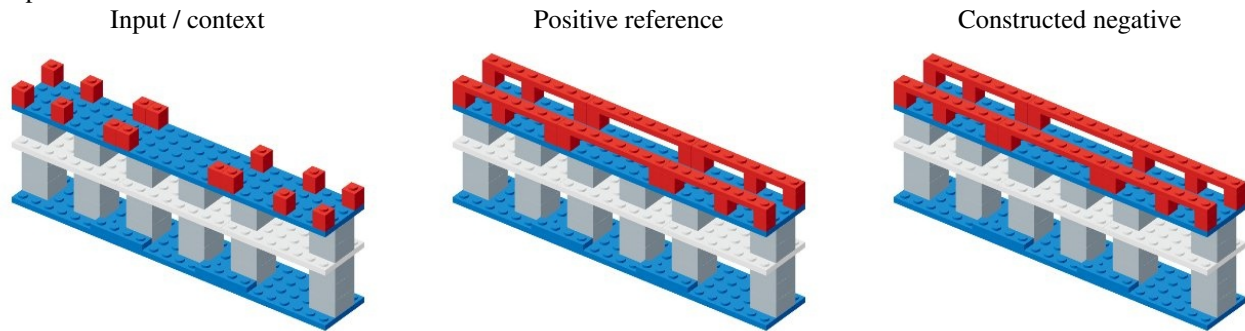


Strict success: positive 1, negative 0. Rejection: selection_mismatch. Machine-readable inputs, both answers and all metrics are in review-gallery/case-33.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

```
Positive: {"legalIds": ["parapet-post-1", "parapet-post-2", "parapet-post-3", "parapet-post-4", "parapet-post-5", "parapet-post-6"]}
Negative: {"legalIds": []}
```

pose-estimation. Case e4d1526119eb949e17077c01.

Place the selected parapet post into its target pose from the current state and reference views; accept equivalent yaw for its footprint.



Strict success: positive 1, negative 0. Rejection: answer_mismatch. Machine-readable inputs, both answers and all metrics are in review-gallery/case-34.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

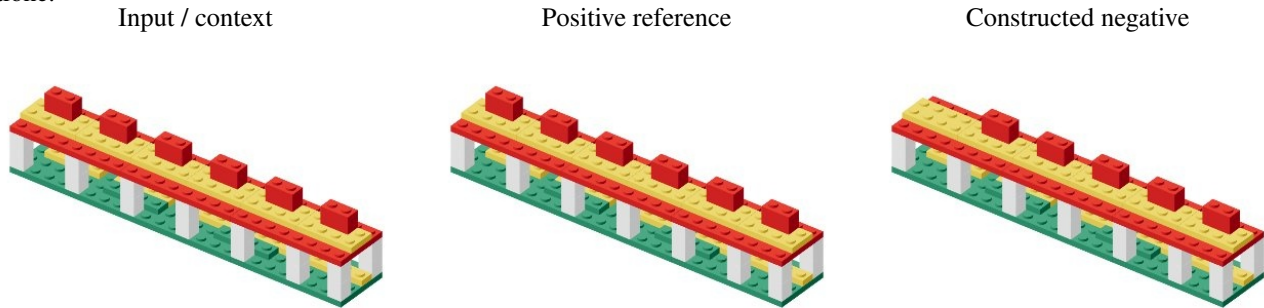
```
Positive: {"x": 15, "y": 18, "z": 0, "turn": 0}
Negative: {"x": 16, "y": 18, "z": 0, "turn": 0}
```

13.18. Covered Market Hall

Source: advanced-covered-market. Two cases share this source. Read numeric/graph answer fields alongside context-only images.

scene-reconstruction. Case 770a5b457843640c815ff8ef.

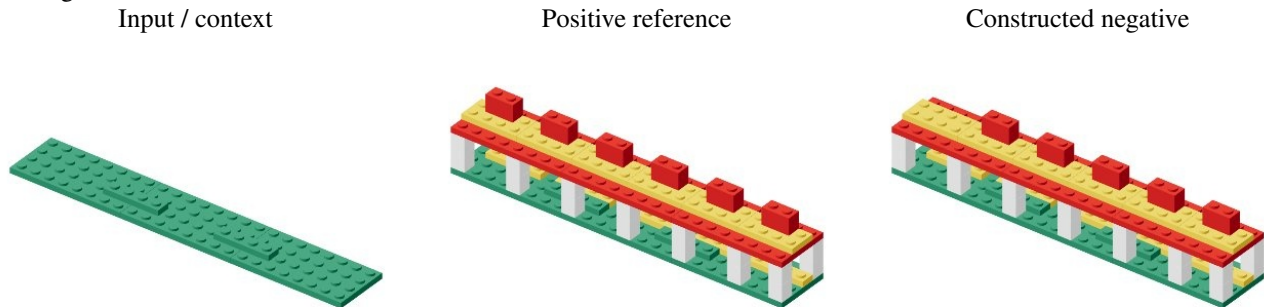
Reconstruct the complete 53-part market from four views and its bill of materials. Repeated columns and signs remain distinct instances. Layer disclosure is supplied: recover complete geometry up to piece yaw symmetry, not exterior RGB alone.



Strict success: positive 1, negative 0. Rejection: target_mismatch. Machine-readable inputs, both answers and all metrics are in review-gallery/case-35.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

constrained-redesign. Case 320d35b94f077a31b364afa0.

Design any connected grid structure preserving the foundation, within inventory and exact extent, with at least four colors and minRoofCells occupied 3D cells among parts whose bottom $Y \geq 7$. This measures a volume proxy, not semantic roof coverage.



Strict success: positive 1, negative 0. Rejection: roofCoverage. Machine-readable inputs, both answers and all metrics are in review-gallery/case-36.json. Field-answer cases use scene images for context only; a matching scene is not a correct field answer.

Object	Parts	Levels	Depth	$X \times Z$
Courtyard Museum	192	7	7	32×24
Intercity Terminal	227	14	14	40×16
Coastal Cargo Vessel	158	11	11	40×14
Four-Tower Citadel	236	11	11	32×24
Process Service Plant	288	16	14	40×24
Four-Storey Archive	408	19	19	32×16

Table 29. Levels count distinct bottom- Y coordinates. Depth is the longest all-lower-overlap precedence chain, not a learned difficulty score.

14. Frontier: Large-Structure Maintenance Diagnostics

Version and scope. Frontier-1 is a separate public-development protocol, not a replacement for frozen v2 or Challenge-2. Six original objects contain 1,509 placed parts and instantiate eight contracts each, yielding 48 tasks. The 117,910 historical conditions are not counted as Frontier tasks. Its nominal geometry extends the envelope to 512 parts, integer origins 0–95 and bodies in $[0, 96]^3$, while retaining the same rectangular catalog and vertical action semantics. No arbitrary CAD connectors, force balance, robot motion, or image-only results are claimed.

14.1. Related-Work Audit and Bounded Opportunity

Generation and larger objects are not novel by themselves: BrickNet [20] supplies much richer human-designed LDraw content and arbitrary connector semantics; BrickGPT [35] includes physics-aware rollback; BrECS [1] already considers part budgets and types. MEPNet [44] handles unseen composed parts and symmetry; TreeSBA [12] predicts assembly from multiple views. BC-Bench [23] directly measures brick selection and pose.

LEGO-Puzzles [37] includes multi-step spatial reasoning and, in its revised submission, a 1–8-step planning set. PhyBlock [24] includes 400 assembly tasks, 2,200 VQA tasks, progressive levels, diagnosis, robustness and counterfactuals. LEGO Co-builder [15] studies fine-grained instruction, object and state detection. Break and Make [43] already couples inspection, disassembly and rebuilding. WorkBenchMark [25] provides 400 Duplo tasks, four tiers and robot/VLA evaluation; Eye-in-Finger [38] addresses occlusion and precise physical error correction.

Thus neither “repair”, “active inspection”, “multiple tasks”, nor “long-horizon assembly” establishes novelty. Our bounded opportunity is a shared-object diagnostic contract tying minimum intervention certificates, executable maintenance, finite-world uncertainty and explicit resource costs together. This combination is less directly covered in the reviewed protocols; we do not prove absence across all

literature or claim a first-ever capability. The Chinese report records primary links, scope and overlaps.

14.2. Eight Executable Contracts

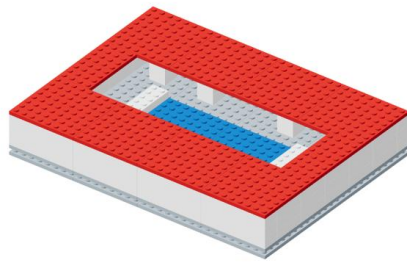
Minimum-action service. Input is a current assembly, full target and service ID. Actions remove a current ID or place that target inventory part at its target pose. No in-place recoloring or group moves are allowed. A legal trace must restore all target parts within the declared budget. Let C be the transitive closure of overhead footprint overlaps starting from service IDs. Every member of C must be removed and replaced, so $2|C|$ is a lower bound. Reverse-height removal followed by height-ordered placement achieves it in these cases, establishing 10–50-action optimal witnesses. This is polynomial closure reasoning, not evidence of general search hardness. Report legal prefix, final exactness, action count, lower bound and excess.

Access certificate. Return exactly C , including service IDs, and any legal removal order. The evaluator checks minimum-set equality and executes the supplied order. An independent upward-graph traversal verifies closure. The IDs removed by the service solver must equal the certificate, providing a same-object cross-task consistency control.

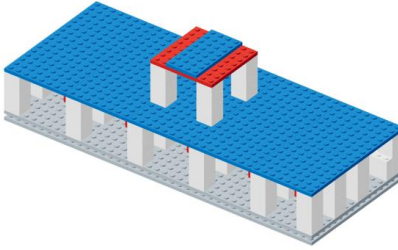
Parallel schedule. Return synchronous batches with at most four abstract workers. Every lower footprint-overlapping part is a prerequisite and must occur in an earlier batch. IDs must occur exactly once; all must finish. Report coverage, makespan and the workload lower bound $\lceil n/4 \rceil$. This conservative all-prerequisite contract is stronger than the one-stud-support predicate. Optimality and simultaneous robot motions are not claimed. A serial height-order baseline succeeds, but takes n rounds; success alone does not reward efficiency.

Minimum support intervention. Choose a minimum-cardinality subset of a public candidate set whose forced simultaneous deletion eventually removes the stated goal under iterative zero-direct-support propagation. Report forced IDs and the entire propagated set, including those IDs. Enumerating the candidate power set establishes optimal cardinality; any optimal choice is accepted. Forced deletion ignores access. This measures a graph/geometry counterfactual, not physical collapse.

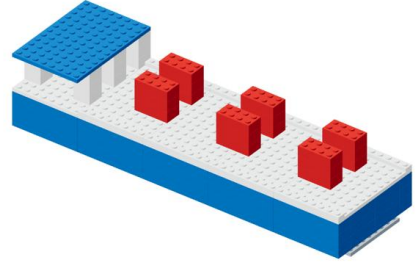
Inventory-constrained replacement. Replace an 8×4 plate using a limited mixed inventory of smaller plates. Local cells must be covered exactly once, without protrusion. The sum of the largest available plate areas yields a cardinality lower bound. Each released instance has a feasible tiling attaining it; an independent public-input exact-cover search



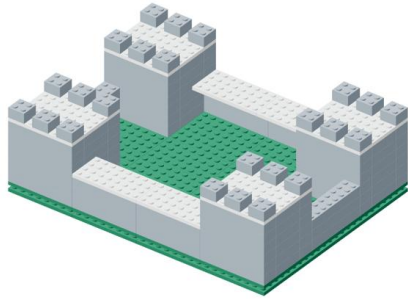
Courtyard Museum
192 parts | 7 levels | depth 7



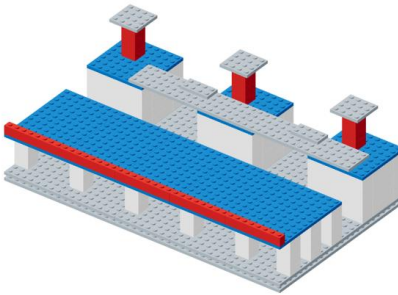
Intercity Terminal
227 parts | 14 levels | depth 14



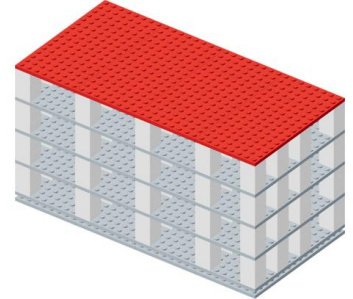
Coastal Cargo Vessel
158 parts | 11 levels | depth 11



Four-Tower Citadel
236 parts | 11 levels | depth 11



Process Service Plant
288 parts | 16 levels | depth 14



Four-Storey Archive
408 parts | 19 levels | depth 19

Figure 8. All six Frontier objects, without selection. Functional colors, deep footprints, hollow towers, service galleries and stacked floors replace the previous four-stud-deep challenge strips. Greater part count is not itself evidence of greater reasoning difficulty.

verifies a witness. Alternative optimal tilings are accepted. The equivalence is occupied slab volume, not unchanged seams, contacts or BOM. These small local problems are calibration tasks, not general semantic redesign.

Admissible hidden topology. Three foundation slabs each admit two four-brick tilings: two pairs of lengthwise plates or four crosswise plates. Their Cartesian product gives eight worlds with equal colored occupied cells and BOM but different seams. Geometry outside the slots, the finite grammar and one slot observation are public. Return all four compatible world IDs. Guessing one private reference is false certainty. Every world is geometry-validated; occupancy equality is not a claim of identical RGB seam pixels. There is no seam RGB input in this track.

Budgeted contingent inspection. The same eight hidden worlds are identified by a query decision tree. Three local queries cost 1, 2 and 3; a full query costs 9. At each internal node, the simulator returns that query’s symbolic value for the current world; the branch determines the next node. Every world must reach its correct leaf with worst-case cost at most 6. Repeated queries, unknown queries, missing branches and incorrect leaves fail. Report world coverage

and worst-case cost. A minimax public-input solver selects queries; no real camera or robot interface is asserted.

Distributed mixed-fault patch. Three regions contain translation, quarter-turn and color faults. The full current and target structures are public. Return exact faulty current IDs and only their replacement part records. Unlisted geometry is preserved, and the reconstructed result must exactly match up to per-part yaw symmetry. The patch is not a physical action trace. This is a transparent diagnostic calibration, not hard perception.

14.3. Baselines, Disclosure and Limits

The public solvers use graph traversal, exhaustive candidate deletion, first-uncovered-cell tiling search, direct reference comparison and minimax query trees. The evaluator and solver do not share target answer objects. Forty-eight positive and forty-eight constructed negative controls separate correctly. These are validator checks, not 96 model predictions. Tests cover schema coercion, coordinate bounds, legal action order, omitted/duplicated steps, alternative tilings, query budgets, all-world coverage and independent cell-map support.

All 48 new tasks are symbolic or finite-observation con-

Task	Public solver	Shortcut
service plan	6/6	0/6
access certificate	6/6	0/6
parallel schedule	6/6	6/6
minimal intervention	6/6	0/6
inventory cover	6/6	0/6
ambiguity set	6/6	0/6
inspection policy	6/6	0/6
distributed repair	6/6	0/6

Table 30. Actual deterministic algorithm runs, six source objects per task. No paid inference or learned-model results. Solvers receive only exported public fields. Shortcut definitions and per-case receipts are released; serial scheduling is valid but slower.

tracts. Four-view, layer and exploded images are reviewer assets, not undeclared model observations. Existing Challenge visual contracts remain separate. Within a track, models receive identical public JSON and budgets; raw and harness-assisted outputs must be reported separately. Use source-grouped statistics on six objects, not 48 independent samples. Model inference, human calibration and externally reproduced results remain unmeasured. Public development data are not contamination-resistant confirmation data. Reference checking and larger scenes do not justify raising empirical discrimination or application-validity claims.

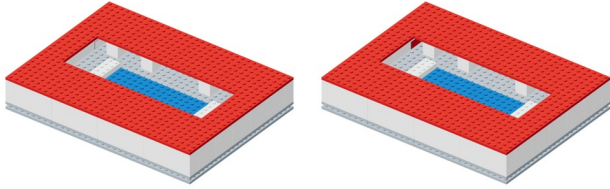
15. Frontier Model and Case Atlas

All 48 pairs are constructed evaluator controls, not model predictions. Field-answer images are context only. Only the exported public JSON is allowed evidence; the plates do not define an image-only track.

15.1. Courtyard Museum

192 parts; 7 bottom-Y levels; dependency depth 7.

service plan

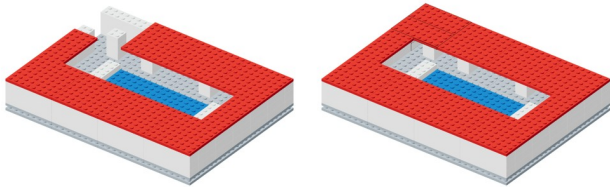


Positive success: 1. Negative success: 0.

Rejection: blocked.

Positive metrics: actions=10, lowerBound=10, finalExact=1, legal-Prefix=1, excessActions=0.

access certificate

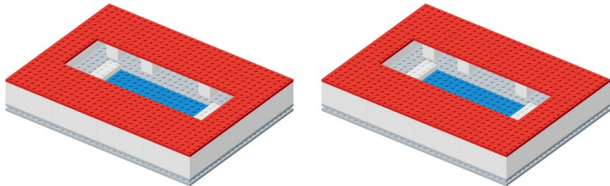


Positive success: 1. Negative success: 0.

Rejection: minimum_set.

Positive metrics: removals=5.

parallel schedule

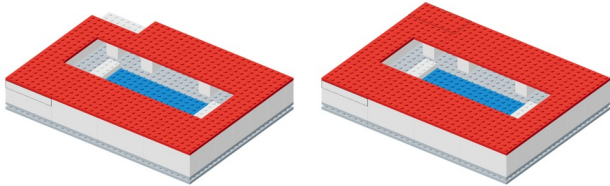


Positive success: 1. Negative success: 0.

Rejection: batch_schema.

Positive metrics: coverage=1, makespan=48, workerLower-Bound=48.

minimal intervention



Positive success: 1. Negative success: 0.

Rejection: goal_survives.

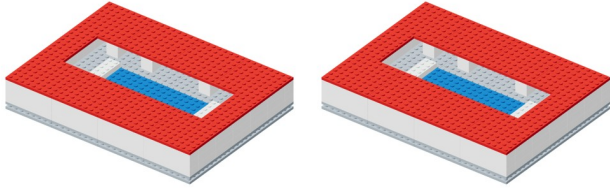
Positive metrics: removals=2, optimum=2.

Positive: {"removeIds": ["p0123", "p0131"], "cascadeIds": ["p0123", "p0131", "p0165"]}

15.2. Courtyard Museum (continued)

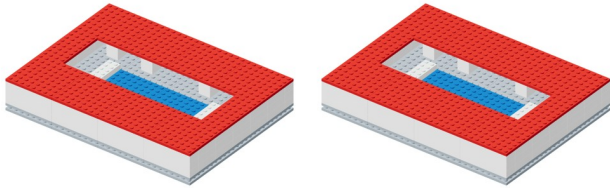
192 parts; 7 bottom-Y levels; dependency depth 7.

inventory cover



Positive success: 1. Negative success: 0.
Rejection: uncovered.
Positive metrics: coverage=1, tiles=7, lowerBound=7.

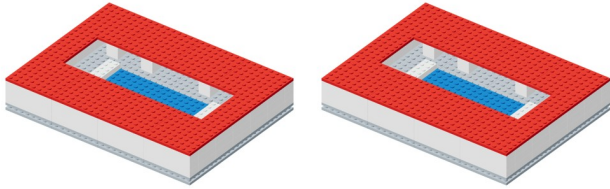
ambiguity set



Positive success: 1. Negative success: 0.
Rejection: admissible_set.
Positive metrics: admissible=4, falseCertainty=0.

Positive: {"possibleIds": ["h0", "h2", "h4", "h6"]}

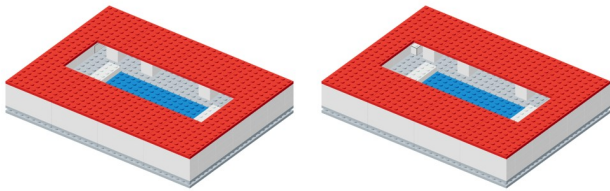
inspection policy



Positive success: 1. Negative success: 0.
Rejection: policy_coverage_or_budget.
Positive metrics: worldCoverage=1, worstCost=6.

Positive policy queries q0, q1, q2 and identifies all eight worlds at worst cost 6; the negative guesses h0 without querying. Full branch tree is in the corresponding review JSON.

distributed repair

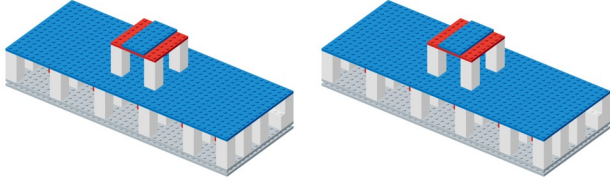


Positive success: 1. Negative success: 0.
Rejection: fault_set.
Positive metrics: finalExact=1, changedRecall=1.

15.3. Intercity Terminal

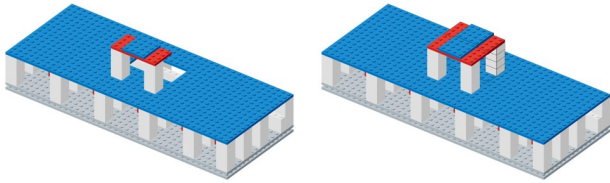
227 parts; 14 bottom-Y levels; dependency depth 14.

service plan



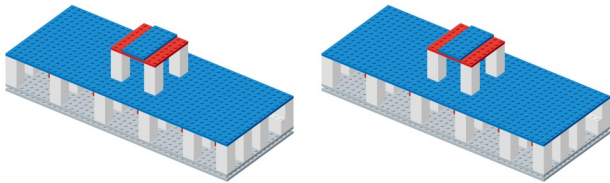
Positive success: 1. Negative success: 0.
Rejection: blocked.
Positive metrics: actions=30, lowerBound=30, finalExact=1, legal-Prefix=1, excessActions=0.

access certificate



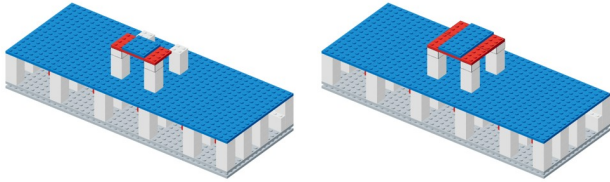
Positive success: 1. Negative success: 0.
Rejection: minimum_set.
Positive metrics: removals=15.

parallel schedule



Positive success: 1. Negative success: 0.
Rejection: batch_schema.
Positive metrics: coverage=1, makespan=58, workerLower-Bound=57.

minimal intervention



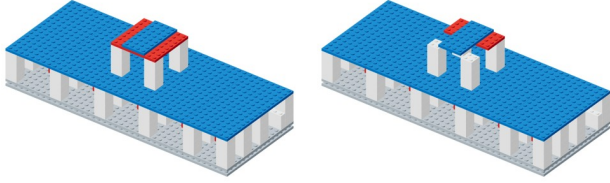
Positive success: 1. Negative success: 0.
Rejection: goal_survives.
Positive metrics: removals=2, optimum=2.

Positive: {"removeIds": ["p0211", "p0219"], "cascadeIds": ["p0211", "p0219", "p0224", "p0226"]}

15.4. Intercity Terminal (continued)

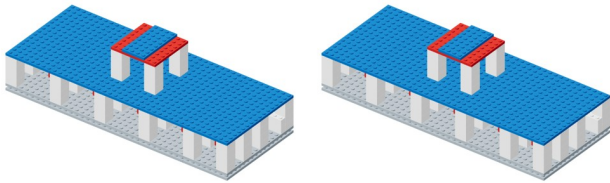
227 parts; 14 bottom-Y levels; dependency depth 14.

inventory cover



Positive success: 1. Negative success: 0.
Rejection: uncovered.
Positive metrics: coverage=1, tiles=5, lowerBound=5.

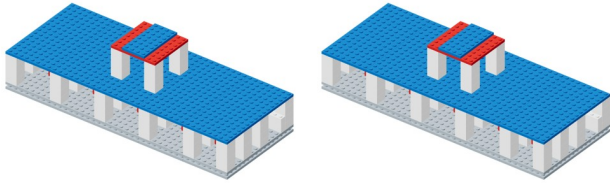
ambiguity set



Positive success: 1. Negative success: 0.
Rejection: admissible_set.
Positive metrics: admissible=4, falseCertainty=0.

Positive: {"possibleIds": ["h0", "h2", "h4", "h6"]}

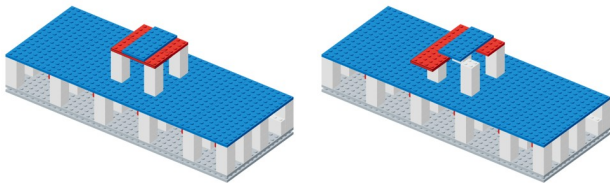
inspection policy



Positive success: 1. Negative success: 0.
Rejection: policy_coverage_or_budget.
Positive metrics: worldCoverage=1, worstCost=6.

Positive policy queries q0, q1, q2 and identifies all eight worlds at worst cost 6; the negative guesses h0 without querying. Full branch tree is in the corresponding review JSON.

distributed repair

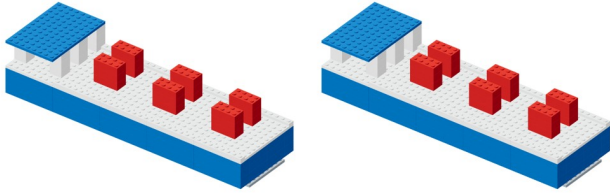


Positive success: 1. Negative success: 0.
Rejection: fault_set.
Positive metrics: finalExact=1, changedRecall=1.

15.5. Coastal Cargo Vessel

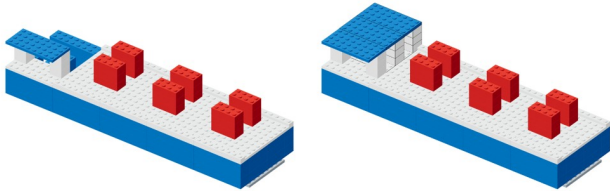
158 parts; 11 bottom-Y levels; dependency depth 11.

service plan



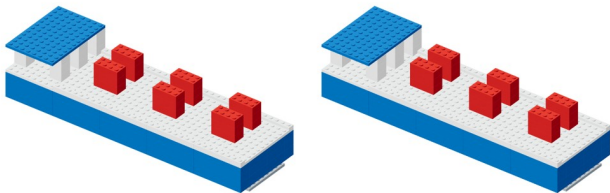
Positive success: 1. Negative success: 0.
Rejection: blocked.
Positive metrics: actions=38, lowerBound=38, finalExact=1, legal-Prefix=1, excessActions=0.

access certificate



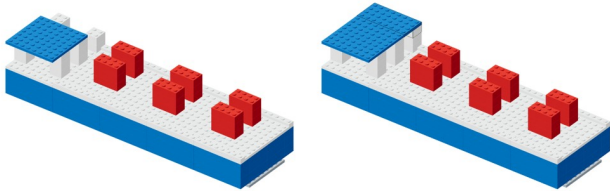
Positive success: 1. Negative success: 0.
Rejection: minimum_set.
Positive metrics: removals=19.

parallel schedule



Positive success: 1. Negative success: 0.
Rejection: batch_schema.
Positive metrics: coverage=1, makespan=40, workerLower-Bound=40.

minimal intervention



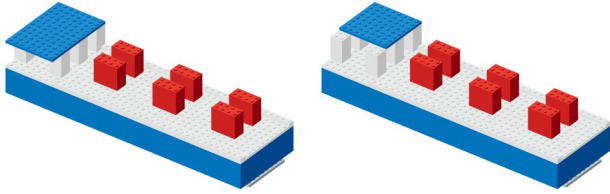
Positive success: 1. Negative success: 0.
Rejection: goal_survives.
Positive metrics: removals=2, optimum=2.

Positive: {"removeIds": ["p0114", "p0123"], "cascadeIds": ["p0114", "p0123", "p0130"]}

15.6. Coastal Cargo Vessel (continued)

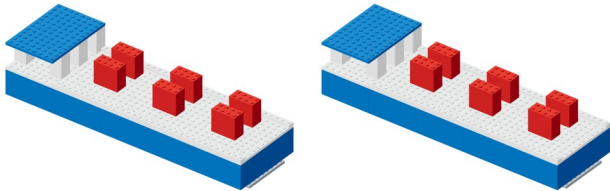
158 parts; 11 bottom-Y levels; dependency depth 11.

inventory cover



Positive success: 1. Negative success: 0.
Rejection: uncovered.
Positive metrics: coverage=1, tiles=5, lowerBound=5.

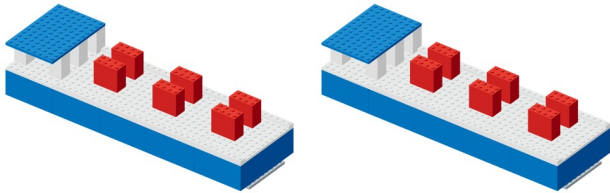
ambiguity set



Positive success: 1. Negative success: 0.
Rejection: admissible_set.
Positive metrics: admissible=4, falseCertainty=0.

Positive: {"possibleIds": ["h0", "h2", "h4", "h6"]}

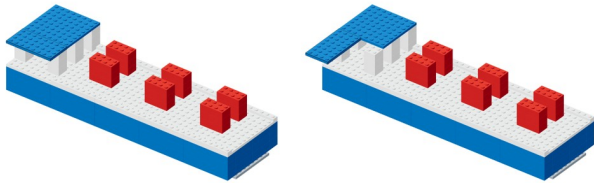
inspection policy



Positive success: 1. Negative success: 0.
Rejection: policy_coverage_or_budget.
Positive metrics: worldCoverage=1, worstCost=6.

Positive policy queries q0, q1, q2 and identifies all eight worlds at worst cost 6; the negative guesses h0 without querying. Full branch tree is in the corresponding review JSON.

distributed repair

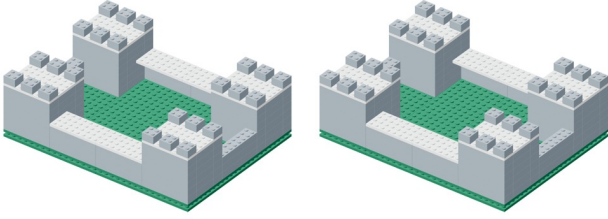


Positive success: 1. Negative success: 0.
Rejection: fault_set.
Positive metrics: finalExact=1, changedRecall=1.

15.7. Four-Tower Citadel

236 parts; 11 bottom-Y levels; dependency depth 11.

service plan

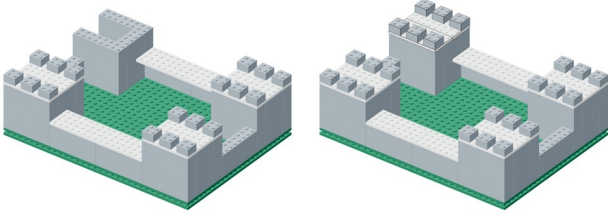


Positive success: 1. Negative success: 0.

Rejection: blocked.

Positive metrics: actions=28, lowerBound=28, finalExact=1, legal-Prefix=1, excessActions=0.

access certificate

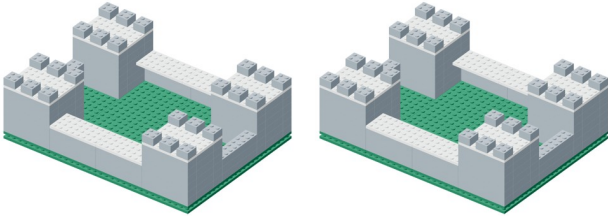


Positive success: 1. Negative success: 0.

Rejection: minimum_set.

Positive metrics: removals=14.

parallel schedule

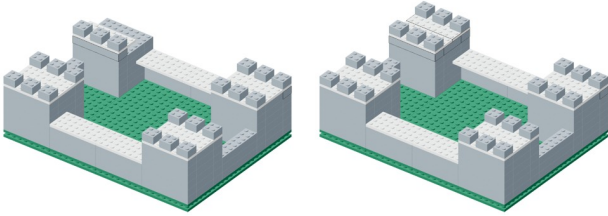


Positive success: 1. Negative success: 0.

Rejection: batch_schema.

Positive metrics: coverage=1, makespan=59, workerLower-Bound=59.

minimal intervention



Positive success: 1. Negative success: 0.

Rejection: goal_survives.

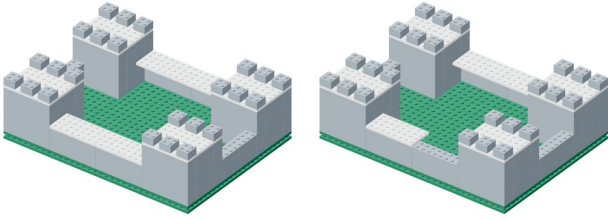
Positive metrics: removals=3, optimum=3.

Positive: {"removeIds": ["p0107", "p0109", "p0110"], "cascadeIds": ["p0107", "p0109", "p0110", "p0111", "p0113", "p0115", "p0117"]}

15.8. Four-Tower Citadel (continued)

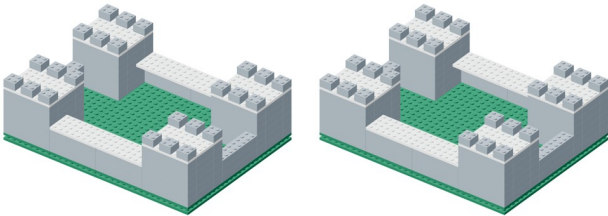
236 parts; 11 bottom-Y levels; dependency depth 11.

inventory cover



Positive success: 1. Negative success: 0.
Rejection: uncovered.
Positive metrics: coverage=1, tiles=5, lowerBound=5.

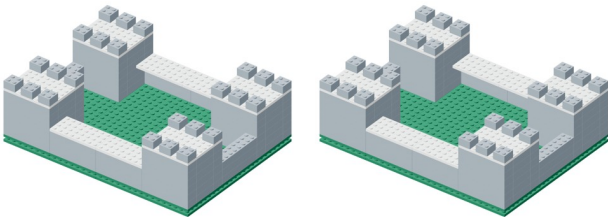
ambiguity set



Positive: {"possibleIds": ["h0", "h2", "h4", "h6"]}

Positive success: 1. Negative success: 0.
Rejection: admissible_set.
Positive metrics: admissible=4, falseCertainty=0.

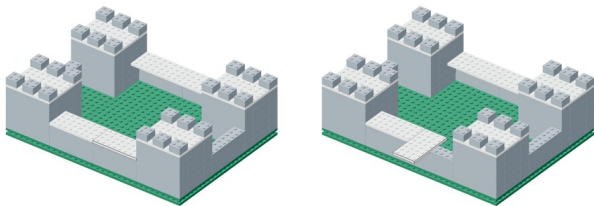
inspection policy



Positive policy queries q0, q1, q2 and identifies all eight worlds at worst cost 6; the negative guesses h0 without querying. Full branch tree is in the corresponding review JSON.

Positive success: 1. Negative success: 0.
Rejection: policy_coverage_or_budget.
Positive metrics: worldCoverage=1, worstCost=6.

distributed repair

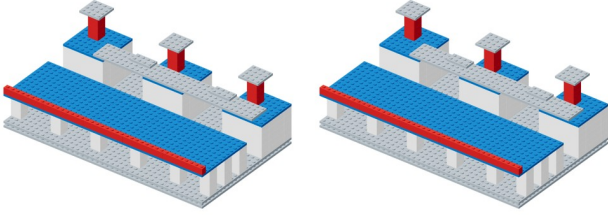


Positive success: 1. Negative success: 0.
Rejection: fault_set.
Positive metrics: finalExact=1, changedRecall=1.

15.9. Process Service Plant

288 parts; 16 bottom-Y levels; dependency depth 14.

service plan

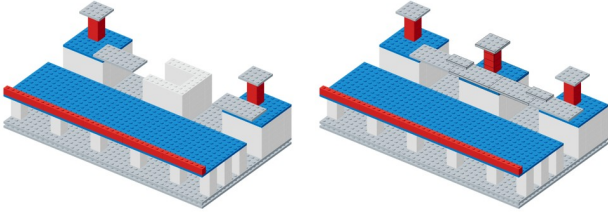


Positive success: 1. Negative success: 0.

Rejection: blocked.

Positive metrics: actions=32, lowerBound=32, finalExact=1, legal-Prefix=1, excessActions=0.

access certificate

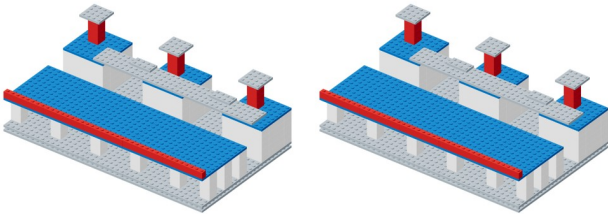


Positive success: 1. Negative success: 0.

Rejection: minimum_set.

Positive metrics: removals=16.

parallel schedule

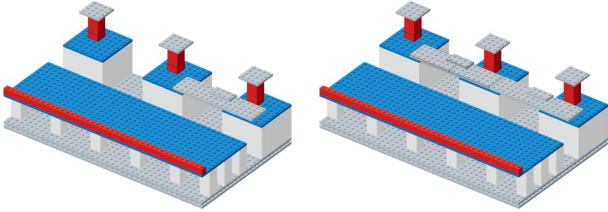


Positive success: 1. Negative success: 0.

Rejection: batch_schema.

Positive metrics: coverage=1, makespan=73, workerLower-Bound=72.

minimal intervention



Positive success: 1. Negative success: 0.

Rejection: goal_survives.

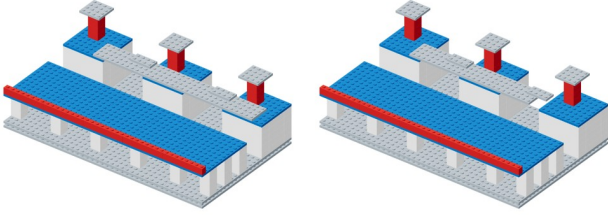
Positive metrics: removals=2, optimum=2.

Positive: {"removeIds": ["p0283", "p0284"], "cascadeIds": ["p0283", "p0284", "p0285"]}

15.10. Process Service Plant (continued)

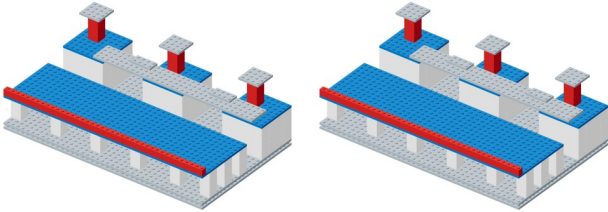
288 parts; 16 bottom-Y levels; dependency depth 14.

inventory cover



Positive success: 1. Negative success: 0.
Rejection: uncovered.
Positive metrics: coverage=1, tiles=7, lowerBound=7.

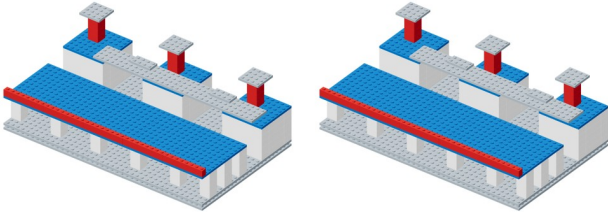
ambiguity set



Positive success: 1. Negative success: 0.
Rejection: admissible_set.
Positive metrics: admissible=4, falseCertainty=0.

Positive: {"possibleIds": ["h0", "h2", "h4", "h6"]}

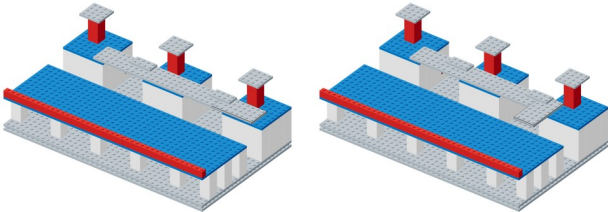
inspection policy



Positive success: 1. Negative success: 0.
Rejection: policy_coverage_or_budget.
Positive metrics: worldCoverage=1, worstCost=6.

Positive policy queries q0, q1, q2 and identifies all eight worlds at worst cost 6; the negative guesses h0 without querying. Full branch tree is in the corresponding review JSON.

distributed repair

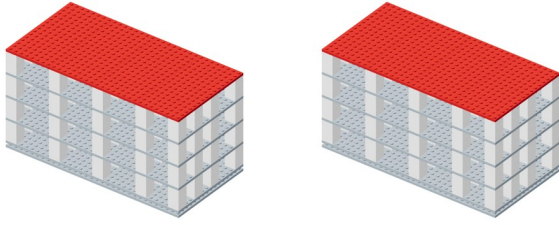


Positive success: 1. Negative success: 0.
Rejection: fault_set.
Positive metrics: finalExact=1, changedRecall=1.

15.11. Four-Storey Archive

408 parts; 19 bottom-Y levels; dependency depth 19.

service plan

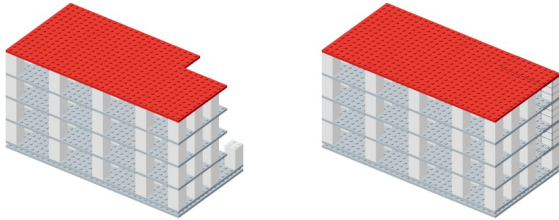


Positive success: 1. Negative success: 0.

Rejection: `blocked`.

Positive metrics: actions=50, lowerBound=50, finalExact=1, legal-Prefix=1, excessActions=0.

access certificate

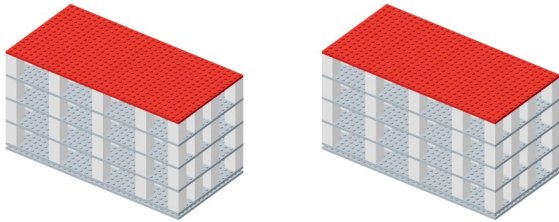


Positive success: 1. Negative success: 0.

Rejection: `minimum_set`.

Positive metrics: removals=25.

parallel schedule

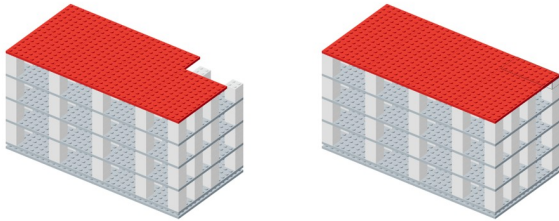


Positive success: 1. Negative success: 0.

Rejection: `batch_schema`.

Positive metrics: coverage=1, makespan=102, workerLower-Bound=102.

minimal intervention



Positive success: 1. Negative success: 0.

Rejection: `goal_survives`.

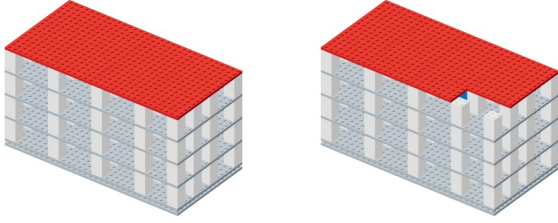
Positive metrics: removals=2, optimum=2.

Positive: `{"removeIds": ["p0359", "p0371"], "cascadeIds": ["p0359", "p0371", "p0405"]}`

15.12. Four-Storey Archive (continued)

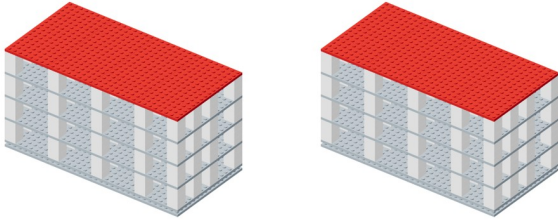
408 parts; 19 bottom-Y levels; dependency depth 19.

inventory cover



Positive success: 1. Negative success: 0.
Rejection: uncovered.
Positive metrics: coverage=1, tiles=7, lowerBound=7.

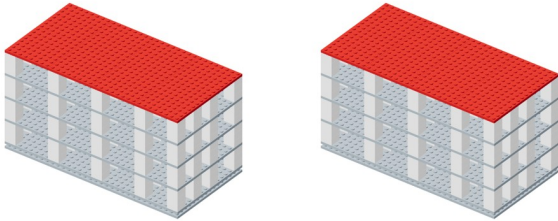
ambiguity set



Positive success: 1. Negative success: 0.
Rejection: admissible_set.
Positive metrics: admissible=4, falseCertainty=0.

Positive: {"possibleIds": ["h0", "h2", "h4", "h6"]}

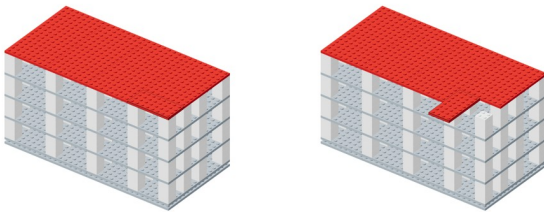
inspection policy



Positive success: 1. Negative success: 0.
Rejection: policy_coverage_or_budget.
Positive metrics: worldCoverage=1, worstCost=6.

Positive policy queries q0, q1, q2 and identifies all eight worlds at worst cost 6; the negative guesses h0 without querying. Full branch tree is in the corresponding review JSON.

distributed repair



Positive success: 1. Negative success: 0.
Rejection: fault_set.
Positive metrics: finalExact=1, changedRecall=1.

16. Diagnostic-2: Revision After Validity Audit

Status and motivation. This section supersedes Frontier-1’s ambiguity and inspection contracts for future evaluation. Historical files and results remain unchanged. An explicit fixed-answer audit showed that all six Frontier ambiguity cases accept the same set, and all six inspection cases accept the same tree. These controls undermine claims of object-dependent reasoning. Diagnostic-2 changes the task mechanism rather than merely renaming IDs.

Consistency evaluation is not new: GQA [16] measures entailed-answer consistency, and Jing et al. [18] study compositional questions and subquestions. Our narrower contract ties fault localization to access closure, inventory-conserving actions and final assembly. It is a measurement proposal with executable controls, not an established new learned-model finding.

16.1. Population and Conditions

There are 138 public conditions from the same six authored sources: 72 maintenance chains (6×3 strata $\times 2$ contexts $\times 2$ reference conditions), 18 admissible-set questions, 18 scan-policy questions, 18 deadline schedules, and 12 mixed-pose patch conditions. The six source objects do not become 138 independent samples. All variants of a source stay in one statistical group.

Maintenance input is the current structure and either full reference JSON or calibrated top images of every bottom- Y layer. Image conditions never include target JSON. Images have a public 20-pixel/stud calibration, origin (60, 100) pixels, positive X right and positive Z down. Reference IDs are not drawn. The current symbolic structure remains privileged context, so this is not pure image-only reconstruction. Do not give models the entire public casebank: another condition of the same source contains privileged reference information. Export one row and its authorized images into an isolated model request.

Maintenance faults are color replacements, deliberately preserving initial geometry while varying access burden. A separate mixed-pose track corrupts translation, yaw and color and asks for a structural patch; it does not claim a valid physical trajectory from an invalid initial state. Equivalent rectangular yaw is accepted; addressed IDs must remain correct. Its learned and specialized visual results are unrun.

16.2. Matched Factors

Three maintenance strata select different buried targets and fault counts. Every configuration has a local context containing its access closure and recursive supporters, and a full context containing the whole object. Both have *exactly the same oracle answer*, checked automatically. Local contexts contain 8–109 parts and full contexts 158–408. This controls output content while changing irrelevant input context. It does not equalize tokenization or total inference cost.

Fault count and selected depth vary together: the strata are not a factorial design. Records expose total parts, access-closure size, fault count, access depth and optimal action count. Actions range from 0 for no-fault controls to 126; maximum access depth is 16. Human difficulty calibration remains absent. These factors support future controlled analyses, not a present assertion of model discrimination.

16.3. Correctness and Coherence

The joint response contains diagnosis D , access certificate A and repair sequence R . Alternatively, 216 stage-specific public views allow three independent requests for each maintenance condition. The scorer merges independent answers without filling missing stages. Joint and independent elicitation have different call budgets and must not be pooled.

Diagnosis returns fault IDs and their corrected parts. Access returns the minimum removal set and an executable order. Repair removes IDs into inventory and places them back, optionally supplying a replacement part with the same ID. It cannot create new instances, repaint in place, or silently change already placed parts. Every action is checked; unknown IDs, duplicate placements and invalid intermediate states fail. All original IDs and exact target geometry must be restored.

Let $C(D)$ be the overhead closure of predicted fault IDs and $M(R)$ the actual removed-ID set. Coherence checks that the declared access set equals $C(D)$ and $M(R)$, and that predicted replacements agree with the final placements. It also requires legal execution. This is distinct from correct diagnosis or final target correctness. The primary joint endpoint requires exact diagnosis, exact access, minimum-action successful repair, and coherence.

A constructed no-operation triple can be coherent but incorrect. A constructed correct repair with an empty diagnosis can be executable and correct at the end but incoherent. Both cases are tested and published. They are evaluator controls, not observations of model behavior. Per-stage accuracy, coherence and the joint endpoint must all be shown.

16.4. Geometry-Dependent Observation

Two, three or four foundation slots admit alternative plate tilings. The grammar contains 4, 6 or 16 candidate worlds; the six-world stratum includes correlated assignment restrictions. All worlds are explicitly valid, but equal occupancy/BOM is not asserted to imply identical seam RGB. This track supplies geometry and symbolic section scans.

A query specifies axis, bottom height, perpendicular coordinate, interval and integer cost. Scanning visits integer cells and reports ordered contiguous runs of the same brick or empty space. The return does not expose part IDs or assignment names. Wide scans may intersect several slots; their information depends on actual spatial alignment. No

precomputed query-to-world return table is present in public input. Candidate and query ordering do not affect valid answers.

Admissible-set tasks receive one observation and return every compatible world. Policy tasks receive none and must identify all worlds within a worst-case cost budget. A minimax enumerator computes a witness from public geometry; every leaf path is checked across every world. Repeated/unknown queries, missing branches and over-budget paths fail. The separate `ScanEpisode` interface returns a query observation before accepting a final guess; a lucky episode is not all-world success.

Among the 18 released optimal answer sets and trees, the best constant transfer succeeds on 2/18 and 3/18 cases, respectively. The legacy `h0, h2, h4, h6` set succeeds on 0/18. Shared `h/q` labels are retained, so this result is not obtained simply by randomizing names. These are finite candidate-transfer controls, not proof against every possible shortcut. Geometry removal makes all scan worlds indistinguishable to the public geometric solver; learned ablations remain unmeasured.

16.5. Deadline Scheduling

Two, four and eight-worker conditions require all lower-overlap prerequisites in earlier rounds and at most the stated number of IDs per batch. Unlike Frontier-1, they also enforce a deadline obtained from a feasible list schedule. Serial ordering fails all 18 deadlines; the released witnesses pass all 18. This does not certify optimal makespan or simultaneous robot execution. Legality, coverage, makespan and meeting the deadline are separate outputs.

16.6. Visual Signal Control and Remaining Gates

A deliberately simple known-renderer control reads public layer PNG pixels and current geometry, recovers reference colors, and computes the maintenance chain. It reads no private target files and succeeds on all 36 image maintenance cases. This demonstrates usable image information under the disclosed calibration, not general perception, pose reconstruction or learned-model capability. Mixed-pose image experiments remain blank.

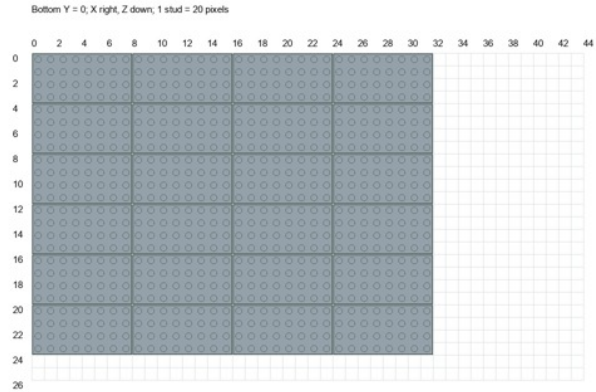
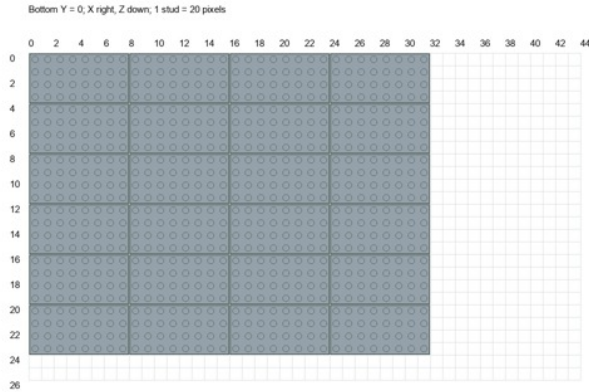
The release freezes public-payload hashes before new learned-model inference. Raw versus tool-assisted answers, modalities and joint versus independent elicitation require separate reporting. Unknown and duplicate IDs are rejected; missing and malformed answers stay in applicable denominators. The prospective human-review table is empty. At least two actual independent reviewers, new external source designs and matched-budget learned-model experiments are still required for scientific validation. Public development sources cannot be relabeled as a hidden confirmation set.

17. Diagnostic-2 Matched Visual Examples

These examples disclose all bottom-Y layers. They are not ordinary exterior-only observations. Each row pairs a current layer with its correct reference at identical grid calibration; other layers remain in the exported input.

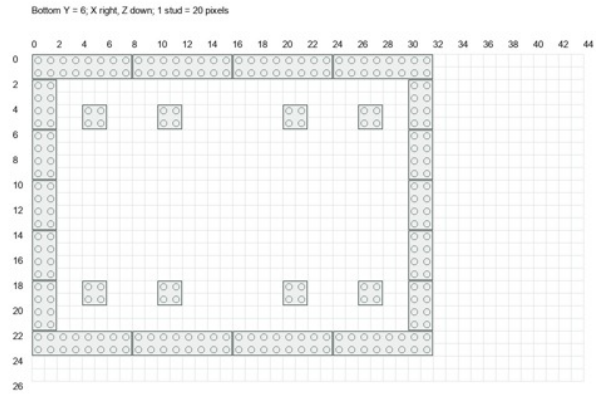
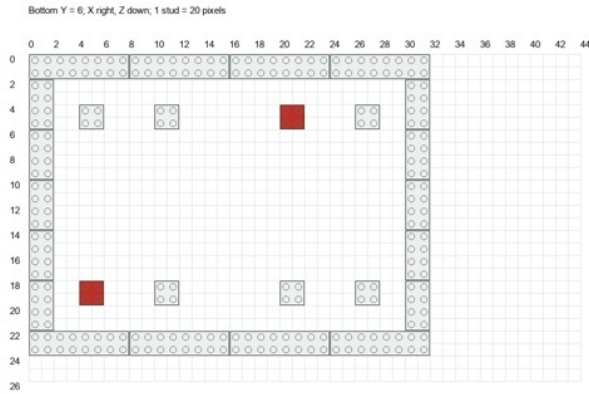
17.1. courtyard museum

Maintenance tier 0



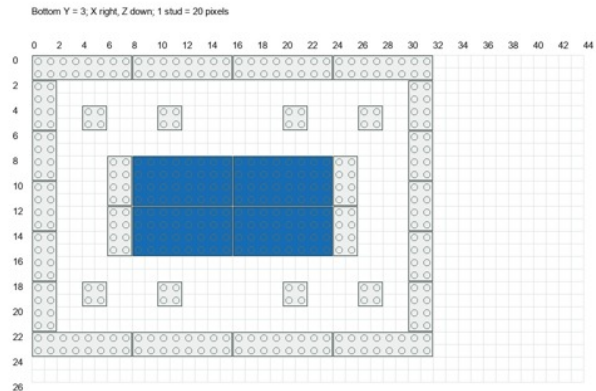
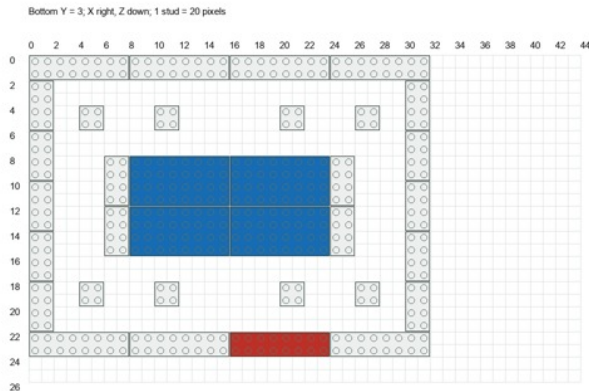
Left: current. Right: reference. Faults: 0; minimum actions: 0. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

Maintenance tier 1



Left: current. Right: reference. Faults: 2; minimum actions: 12. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

Maintenance tier 2

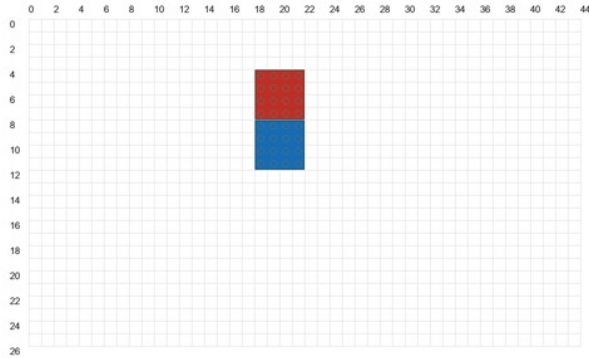


Left: current. Right: reference. Faults: 3; minimum actions: 24. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

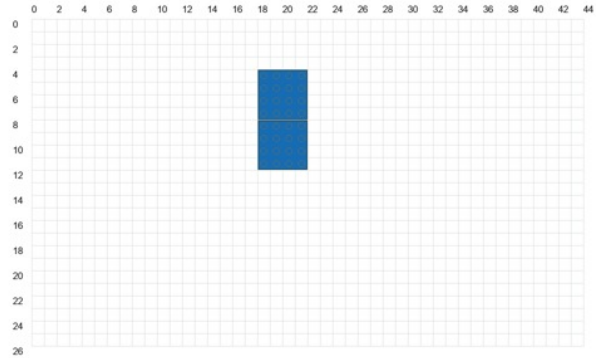
17.2. intercity terminal

Maintenance tier 0

Bottom Y = 29; X right, Z down; 1 stud = 20 pixels



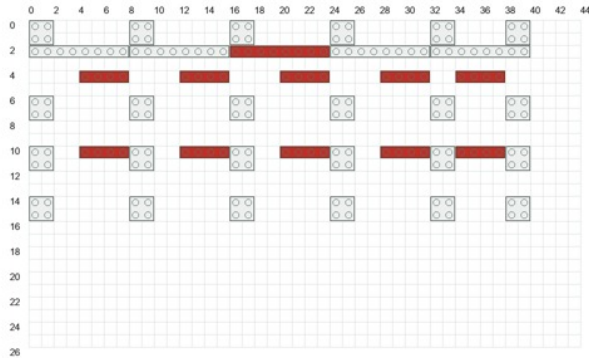
Bottom Y = 29; X right, Z down; 1 stud = 20 pixels



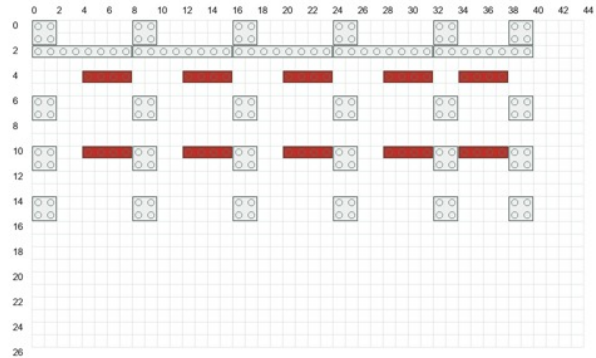
Left: current. Right: reference. Faults: 1; minimum actions: 2. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

Maintenance tier 1

Bottom Y = 3; X right, Z down; 1 stud = 20 pixels



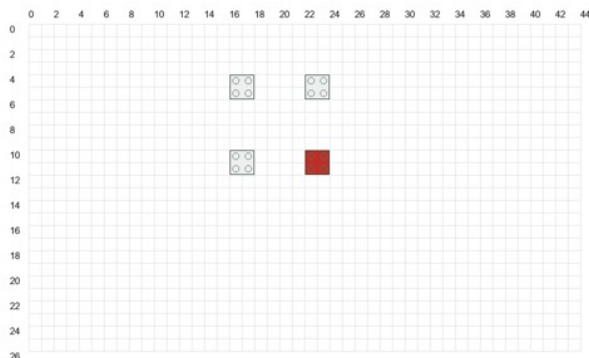
Bottom Y = 3; X right, Z down; 1 stud = 20 pixels



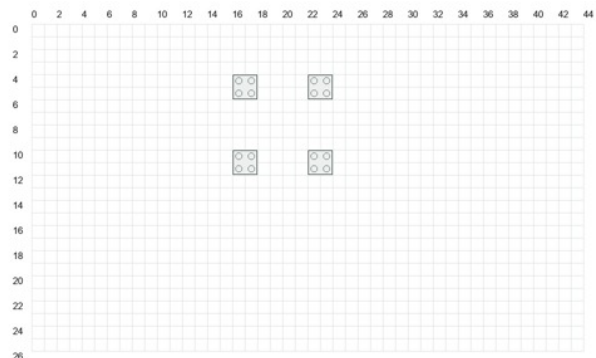
Left: current. Right: reference. Faults: 2; minimum actions: 12. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

Maintenance tier 2

Bottom Y = 19; X right, Z down; 1 stud = 20 pixels

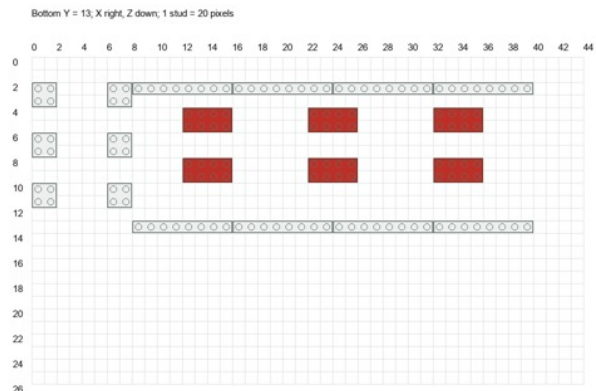


Bottom Y = 19; X right, Z down; 1 stud = 20 pixels

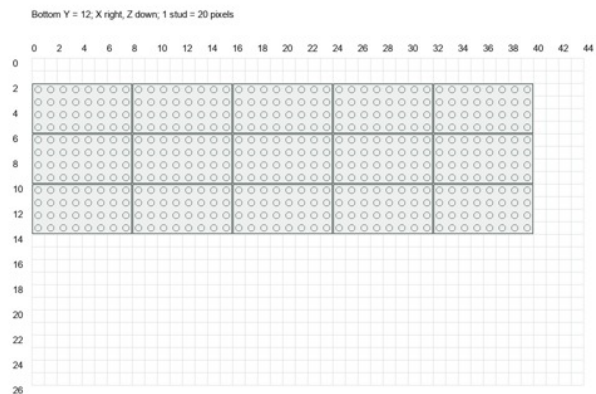


Left: current. Right: reference. Faults: 3; minimum actions: 30. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

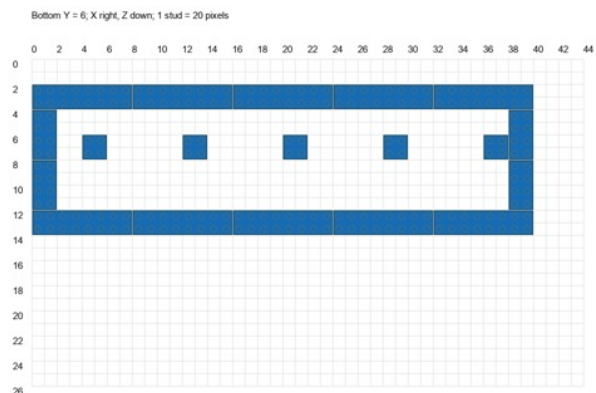
Maintenance tier 0



Maintenance tier 1



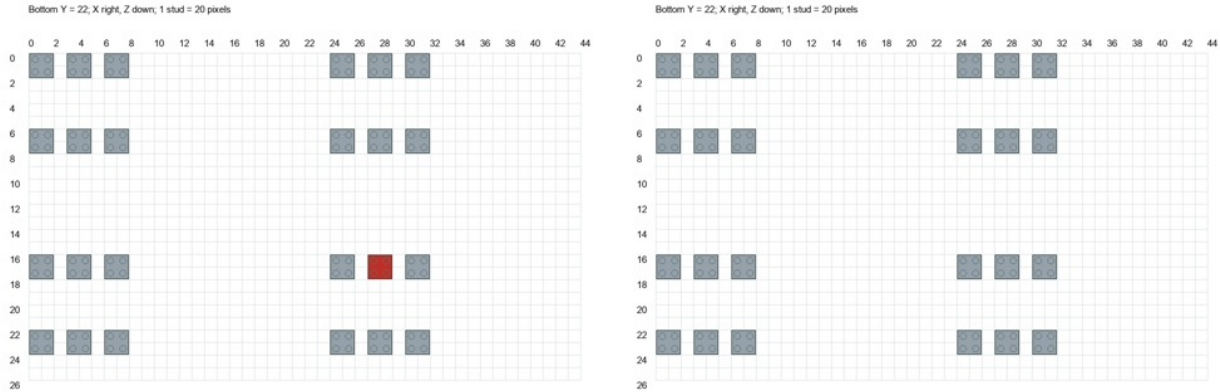
Maintenance tier 2



63

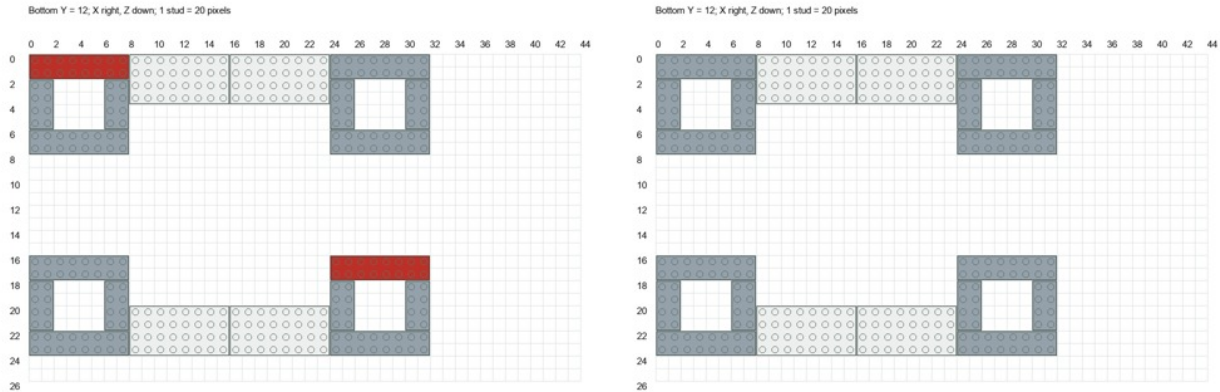
17.4. four tower citadel

Maintenance tier 0



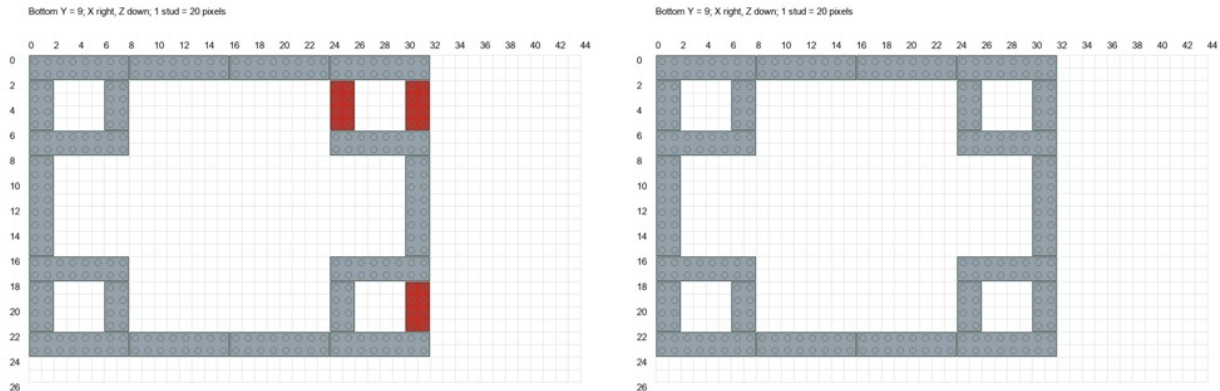
Left: current. Right: reference. Faults: 1; minimum actions: 2. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

Maintenance tier 1



Left: current. Right: reference. Faults: 2; minimum actions: 28. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

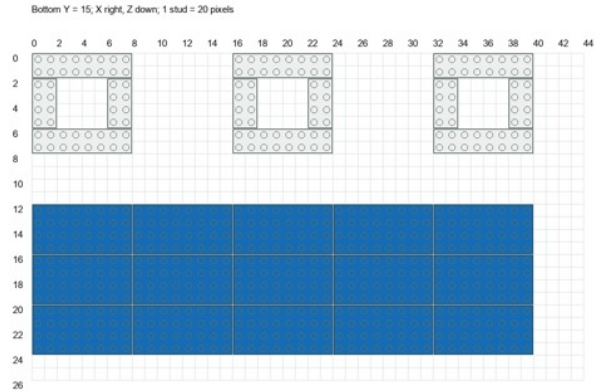
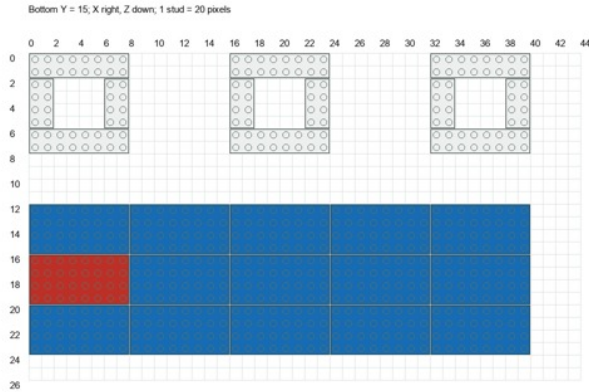
Maintenance tier 2



Left: current. Right: reference. Faults: 3; minimum actions: 56. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

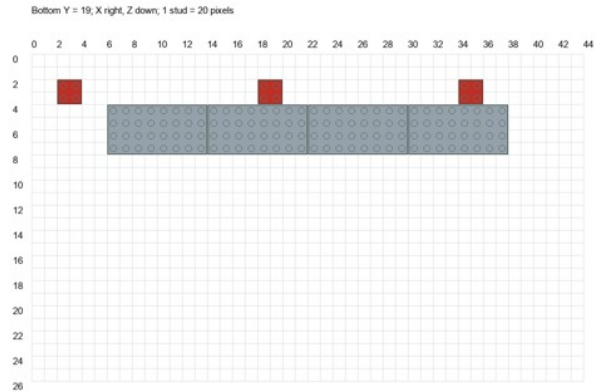
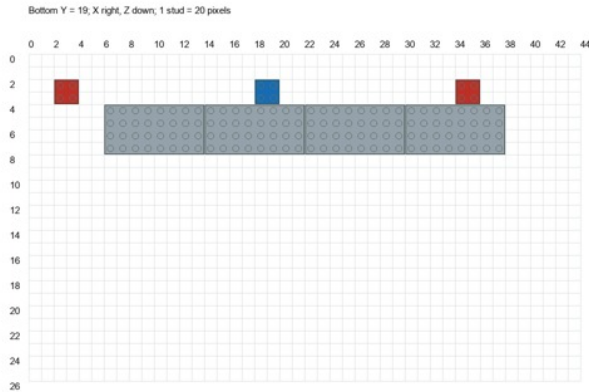
17.5. process service plant

Maintenance tier 0



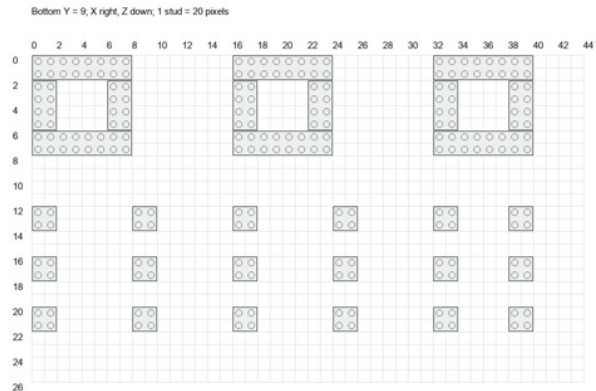
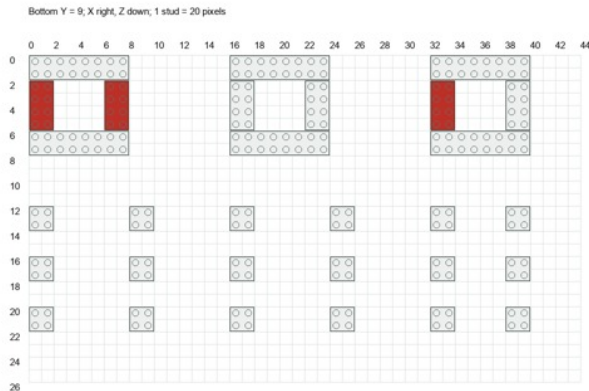
Left: current. Right: reference. Faults: 1; minimum actions: 2. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

Maintenance tier 1



Left: current. Right: reference. Faults: 2; minimum actions: 18. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

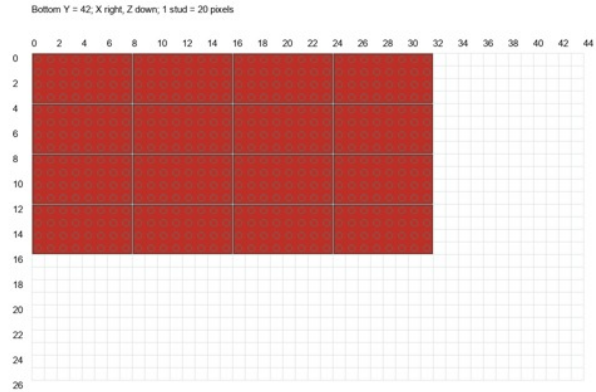
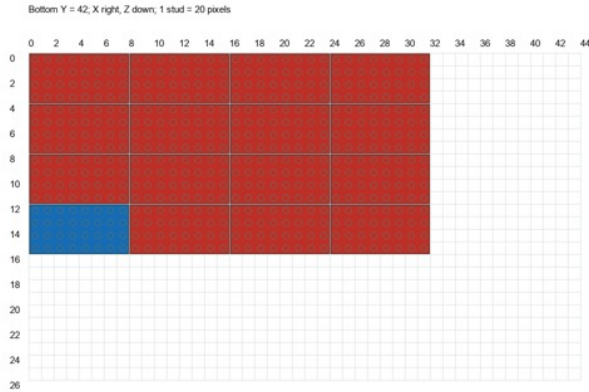
Maintenance tier 2



Left: current. Right: reference. Faults: 3; minimum actions: 54. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

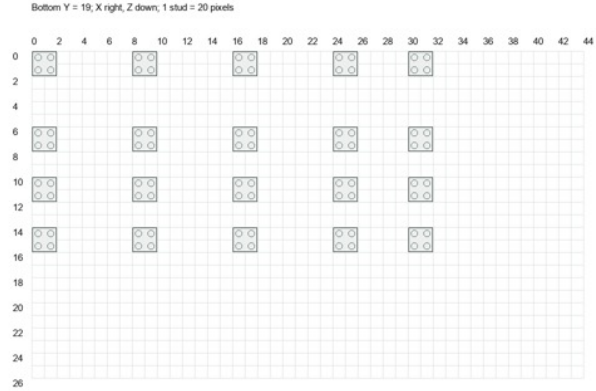
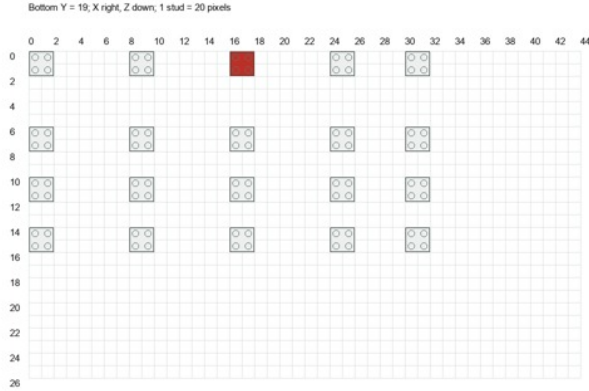
17.6. four storey archive

Maintenance tier 0



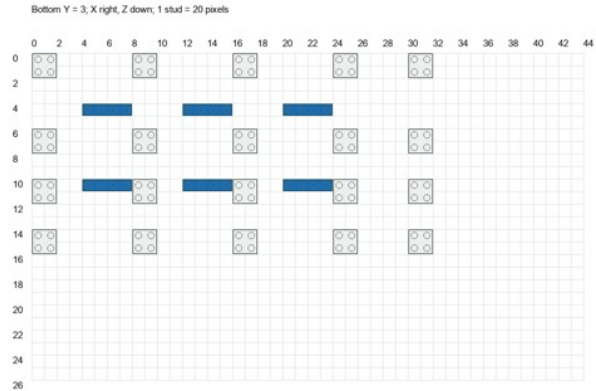
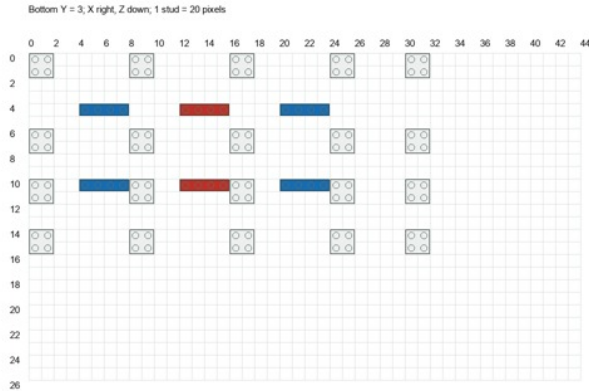
Left: current. Right: reference. Faults: 1; minimum actions: 2. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

Maintenance tier 1



Left: current. Right: reference. Faults: 2; minimum actions: 40. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

Maintenance tier 2



Left: current. Right: reference. Faults: 3; minimum actions: 126. The local/full counterpart has exactly the same answer. All stages use the same reference disclosure.

References

- [1] Seokjun Ahn, Jungtaek Kim, Minsu Cho, and Jaesik Park. Budget-aware sequential brick assembly with efficient constraint satisfaction. *Transactions on Machine Learning Research*, 2024.
- [2] Anthropic. Introducing claude haiku 4.5. <https://www.anthropic.com/news/claude-haiku-4-5>, 2025.
- [3] Apple Machine Learning Research. MLX: An array framework for apple silicon. <https://github.com/ml-explore/mlx>, 2023.
- [4] Artifex Software. Pymupdf documentation. <https://pymupdf.readthedocs.io/>, 2026.
- [5] Andrei Z. Broder. On the resemblance and containment of documents. In *Compression and Complexity of Sequences*, pages 21–29, 1997.
- [6] Anthony Brohan, Noah Brown, Justice Carbajal, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning*, 2023.
- [7] CVPR Author Kit Contributors. Cvpr author kit. <https://github.com/cvpr-org/author-kit>, 2026. Pinned revision 291758547e923160eb4d37079b7b9f0dfce82355.
- [8] Bradley Efron. Bootstrap methods: Another look at the jack-knife. *The Annals of Statistics*, 7(1):1–26, 1979.
- [9] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021.
- [10] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020.
- [11] Google DeepMind. Gemini models. <https://ai.google.dev/gemini-api/docs/models>, 2026. Living documentation; run-time identifiers and prices are frozen in BrickAtlas receipts.
- [12] Mengqi Guo, Chen Li, Yuyang Zhao, and Gim Hee Lee. TreeSBA: Tree-transformer for self-supervised sequential brick assembly. In *European Conference on Computer Vision (ECCV)*, 2024.
- [13] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. Array programming with NumPy. *Nature*, 585:357–362, 2020.
- [14] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.
- [15] Haochen Huang, Yue Su, Xin Sun, Moonisa Ahsan, Mohammad Aliannejadi, Irene Viola, Zhaochun Ren, Chuang Yu, Aneta Lisowska, Artem Belopolsky, Koen Hindriks, Pablo Cesar, Junxiao Wang, and Jiahuan Pei. LEGO Co-builder: Exploring fine-grained vision-language modeling for multimodal assembly assistants. *arXiv preprint arXiv:2507.05515*, 2025. Version 4 revised July 2026.
- [16] Drew A. Hudson and Christopher D. Manning. GQA: A new dataset for real-world visual reasoning and compositional question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.
- [17] John D. Hunter. Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007.
- [18] Chenchen Jing, Yunde Jia, Yuwei Wu, Xinyu Liu, and Qi Wu. Maintaining reasoning consistency in compositional visual question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.
- [19] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, et al. OpenVLA: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.
- [20] Peter Kulits and Cordelia Schmid. BrickNet: Graph-backed generative brick assembly. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 39252–39261, 2026.
- [21] LDraw.org. Ldraw file format specification. <https://www.ldraw.org/article/218.html>, 2026. Specification reference, not authorization to redistribute third-party models.
- [22] Percy Liang, Rishi Bommasani, Tony Lee, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023.
- [23] Jiateng Liu, Bingxuan Li, Zhenhailong Wang, Rushi Wang, Kaiwen Hong, Cheng Qian, Jiayu Liu, Denghui Zhang, Katherine Driggs-Campbell, Manling Li, and Heng Ji. Brick-composer: MLLMs construct everything from building blocks. *arXiv preprint arXiv:2606.05445*, 2026.
- [24] Liang Ma, Jiajun Wen, Min Lin, Rongtao Xu, Xiwen Liang, Bingqian Lin, Jun Ma, Yongxin Wang, Ziming Wei, Haokun Lin, Mingfei Han, Meng Cao, Bokui Chen, Ivan Laptev, and Xiaodan Liang. PhyBlock: A progressive benchmark for physical understanding and planning via 3d block assembly. In *Advances in Neural Information Processing Systems*, 2025.
- [25] Wenbo Ma, Daniel Swoboda, Matteo Tschesche, and Till Hofmann. WorkBenchMark: A LEGO-based assembly benchmark with an assembly-by-disassembly baseline for the smart manufacturing league. *arXiv preprint arXiv:2606.19358*, 2026.
- [26] Aman Madaan, Niket Tandon, Prakhar Gupta, et al. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, 2023.
- [27] Meta. The llama 4 herd. <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>, 2025.
- [28] Microsoft. Playwright documentation. <https://playwright.dev/docs/intro>, 2026.
- [29] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. In *Conference on Fairness, Accountability, and Transparency*, 2019.
- [30] Node.js and TypeScript Contributors. Node.js and typescript documentation. <https://nodejs.org/docs/latest/api/>; <https://www.typescriptlang.org/docs/>, 2026.

- [31] OpenAI. Introducing GPT-4.1 in the api. <https://openai.com/index/gpt-4-1/>, 2025. Official model announcement; accessed 2026-09-14.
- [32] OpenRouter. Api reference. <https://openrouter.ai/docs/api-reference/overview>, 2026. Accessed 2026-09-14; receipts, not current prices, determine reported cost.
- [33] Adam Paszke, Sam Gross, Francisco Massa, et al. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, 2019.
- [34] Pillow Contributors. Pillow documentation. <https://pillow.readthedocs.io/>, 2026.
- [35] Ava Pun, Kangle Deng, Ruixuan Liu, Deva Ramanan, Changliu Liu, and Jun-Yan Zhu. Generating physically stable and buildable brick structures from text. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14798–14809, 2025.
- [36] Qwen Team. Qwen3-VL: Official repository and model cards. <https://github.com/QwenLM/Qwen3-VL>, 2025.
- [37] Kexian Tang, Junyao Gao, Yanhong Zeng, Haodong Duan, Yanan Sun, Zhening Xing, Wenran Liu, Kaifeng Lyu, and Kai Chen. LEGO-Puzzles: How good are MLLMs at multi-step spatial reasoning? *arXiv preprint arXiv:2503.19990*, 2025. Revised public submission includes a 1–8-step planning set; accessed 2026-09-15.
- [38] Zhenran Tang, Ruixuan Liu, and Changliu Liu. Eye-in-finger: Smart fingers for delicate assembly and disassembly of LEGO. *arXiv preprint arXiv:2503.06848*, 2025.
- [39] Robert Tarjan. Depth-first search and linear graph algorithms. *SIAM Journal on Computing*, 1(2):146–160, 1972.
- [40] The Tectonic Project. Tectonic typesetting system. <https://tectonic-typesetting.github.io/>, 2026.
- [41] Three.js Contributors. Three.js documentation. <https://threejs.org/docs/>, 2026.
- [42] Vite Contributors. Vite documentation. <https://vite.dev/guide/>, 2026.
- [43] Aaron Walsman, Muru Zhang, Klemen Kotar, Karthik Desingh, Ali Farhadi, and Dieter Fox. Break and make: Interactive structural understanding using LEGO bricks. In *European Conference on Computer Vision (ECCV)*, 2022.
- [44] Ruocheng Wang, Yunzhi Zhang, Jiayuan Mao, Chin-Yi Cheng, and Jiajun Wu. Translating a visual LEGO manual to a machine-executable plan. In *European Conference on Computer Vision (ECCV)*, 2022.
- [45] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.
- [46] Thomas Wolf, Lysandre Debut, Victor Sanh, et al. Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP: System Demonstrations*, pages 38–45, 2020.
- [47] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023.